pixelactive

CITYSCAPE 1.8

REAL CITIES REAL FAST

Quick Start Tutorial



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Contents

START HERE 1
Running CityScape 1
Running CityScape for the First Time 1
Loading the Demo Project in the Demo Version 2
Loading the Demo Project in the Full Version 2
Loading the Sample Terrain 4
INTRODUCING THE UI AND CAMERA 5
Overview of the User Interface 5
Main Menu 6
Tools Palette 6
3D Render Window 6
Main Widget Panel 7
Library Browser 8
Using the Camera 8
Perspective Camera 8
Orthographic Camera 10
CREATING AND EDITING WORLDS 11
Using Geography Tools 12
The Geography Circle Tool 12
The Geography Polygon Tool 13
The Geography Grid Tool 16
The Geography Grid Paint Tool 18
The Geography Global Paint Tool 19
Using Road Tools 19
About The Road Style Browser 19
The Road Tool 20
More Information on Road Tools 33
Using Water Tools 33
The Global Water Tool 33
The Water Polygon Tool 33
Using Material Tools 35
About the Material Browser 35
The Material Seed Tool 36
The Global Material Feature 36
Edit Materials 37
Using Prop Tools 37
About The Prop Browser 37
Importing New Props 38
The Prop Point Tool 40
The Prop Circle Tool 41
The Prop Polygon Tool 42
The Prop Range Tool 43
Using Sky Tools 45
The Sky Tool 45
The Clouds Tool 45
Using GIS Tools 46
Create a New Geo-Located Terrain 46
Import Elevation Data 47
Import Roads, Buildings, Water, and Props 47
Import an Orthographic Image 48

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Quick Start Tutorial

START HERE

PixelActive CityScape is an interactive tool for creating large 3D environments for realtime rendering. CityScape has all the capabilities that you need to create and populate 3D environments ranging from wilderness terrains to large detailed cities – for use in video games, training simulations, GPS navigation, urban planning, and many more applications.

This “Quick Start Tutorial” provides a hands-on tour of CityScape to quickly familiarize you with the basics. For best results, follow the topics in the order they are presented, with a new terrain loaded in CityScape in front of you. To begin, follow the instructions below:

- Running CityScape
- Loading the Demo Project in the Full Version

Running CityScape

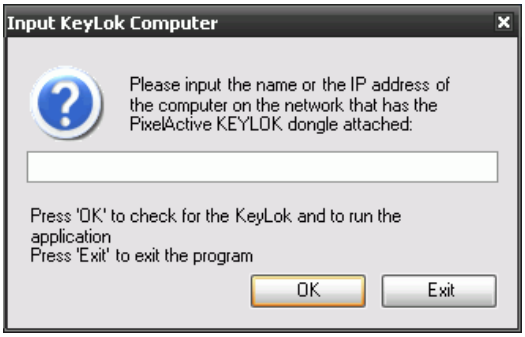
After CityScape is installed, a line entry for CityScape appears in the list of Windows Programs. You can run CityScape by selecting the line entry.

If you have the Demo version of CityScape, skip to Loading the Demo Project in the Demo Version. If you have the Full version, see Running CityScape for the First Time below.

RUNNING CITYSCAPE FOR THE FIRST TIME

If you have the Demo version, licensing is not required. The Full version of CityScape requires that you license your copy of CityScape the first time you run it. Here’s how:

1. The first time you run CityScape, the *Input KeyLok Computer* dialog may appear (see figure below). In this dialog, enter the **network name** (or IP address) of the License Server, and LEFT-CLICK **OK**. (For a standalone installation, with the dongle physically attached to the same machine as the CityScape editor, enter **localhost** in the dialog.).



2. CityScape will execute after it contacts the License Server. When running CityScape in the future, you will not have to re-enter the name of the License Server, unless the server changes.

Loading the Demo Project in the Demo Version

When you run the Demo version of CityScape, a blank terrain automatically loads. This blank terrain is associated with the Demo Project containing various 2D and 3D items (e.g. textured trees, buildings, road styles, etc.) that you can use while doing this Quick Start.

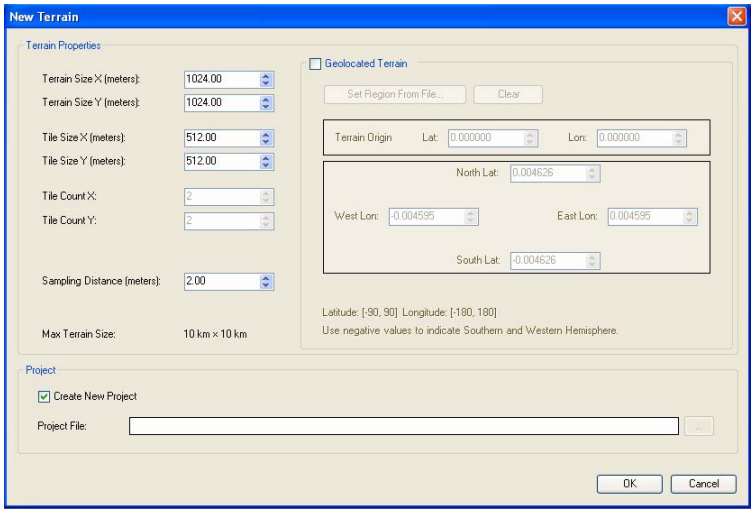
In the Demo version, you can create a new terrain at any time, and associate the new terrain with the Demo project, using the same procedure as the Full version (see the next section).

Loading the Demo Project in the Full Version

When you run the full version of CityScape, you have two options: manually create a new terrain or load an existing terrain. To do this Quick Start with the full version, create a new terrain as follows:

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1. In the Main Menu, choose **File > New Terrain**. The New Terrain dialog appears.



2. Near the bottom of New Terrain dialog, in the Project area, you can choose the Demo project file as follows:
   i. LEFT-CLICK the browse button to the right of the *Project File* field.
   ii. In the file loader that appears, open the **demo.csp** file in the World's folder, which is installed in the following default location for Vista and Win 7 users:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds

or here for WinXP users:

C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds

3. In the New Terrain dialog box, LEFT-CLICK **OK** to give birth to a new terrain.

LOADING THE SAMPLE TERRAIN

If you want to see what a partly constructed terrain looks like, you can open the sample terrain file that comes with CityScape:

1. Select **File / Open Terrain (Ctrl-O)**.
2. In the Open dialog, load the **sample_terrain.cst** sample file, which is in your *Worlds* folder located here by default:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds\Terrains\SampleTerrain

or here for WinXP users:

C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds\Terrains\SampleTerrain

3. After you're done checking out the sample terrain, return to the blank terrain you created in the previous section to continue this Quick Start.



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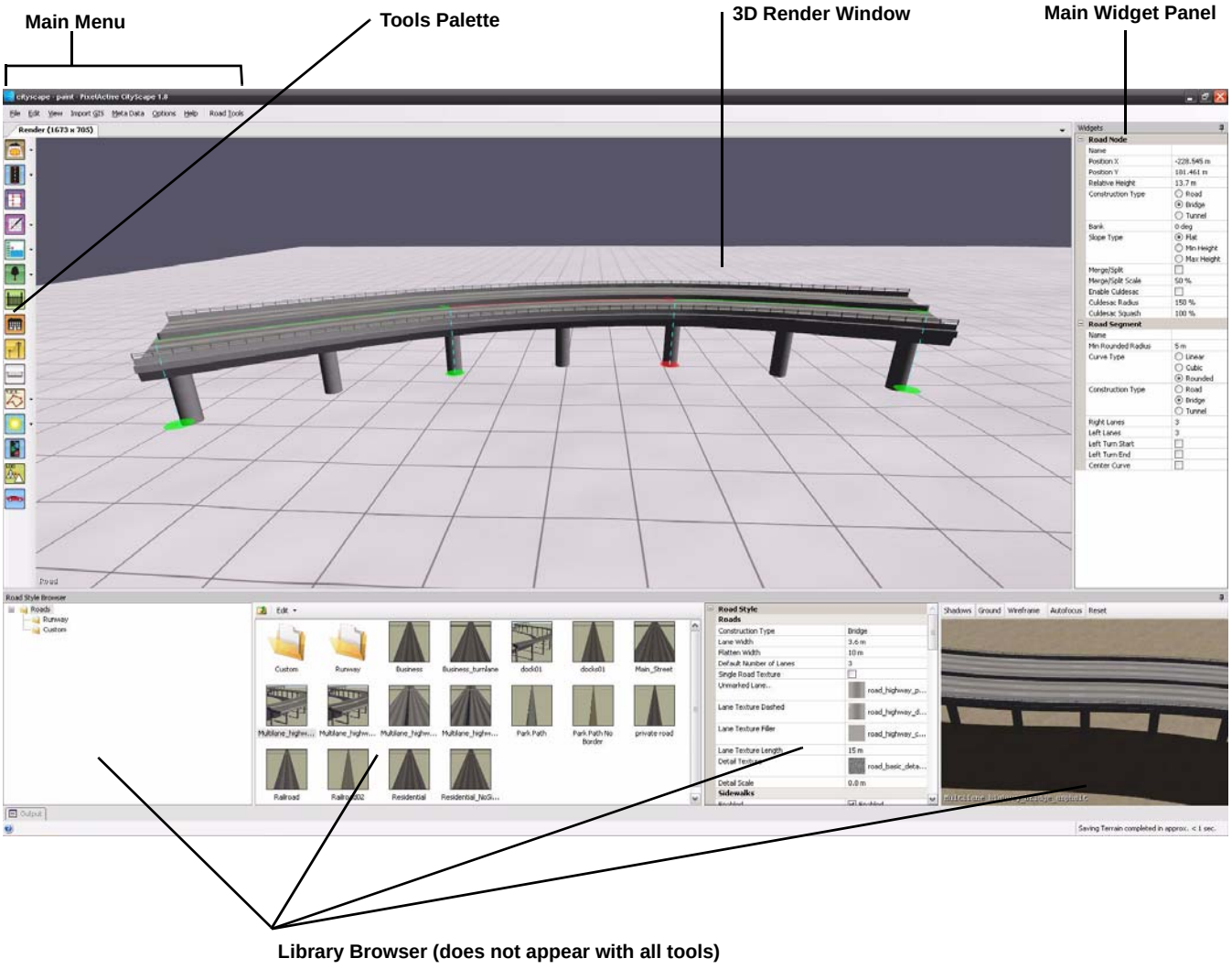
INTRODUCING THE UI AND CAMERA

The sections below introduce the user interface (UI) and camera. (For details on the UI, see “Chapter 3: User Interface” in the *CityScape User Manual*.)

- Overview of the User Interface next.
- “Using the Camera” on page 8

Overview of the User Interface

The following figure provides a quick look at the main elements in CityScape’s User Interface (UI). You can adjust the size of each element by dragging its borders with the mouse. You can also detach each element and move it around. To reset all the UI elements to the default, close CityScape and then hold the **SHIFT-KEY** while restarting CityScape.



Main Menu

The Main Menu provides access to much of the functionality in CityScape. A quick summary is below. For details on each menu, see “Chapter 3: User Interface” in the *CityScape User Manual.*

- **File:** includes the usual new, open, and save terrain commands – as well as export options. (Note: You cannot save or export terrains with the Demo version of CityScape.)
- **Edit:** includes standard cut / copy / paste and undo / redo commands, as well as the Enable Grid Snap option.
- **View:** includes options for cameras, real-time rendering, and visibility toggles (so you can show / hide stuff like overlay images).
- **Import GIS:** If your terrain is geo-located, then you can import a wide range of geo-specific data into the terrain.
- **Meta Data:** allows you to define and assign meta data (i.e. arbitrary ascii data)
- **Options:** collects miscellaneous options such as the ability to change key Terrain Settings (e.g. the size, geo-location, and tile breakdown of the open terrain).
- **Help:** gives you access to this *User Manual* and other help resources as they become available.
- **Road Tools:** When you activate any Road Tool, the Road Tools menu appears to provide you with additional functionality.
- **Water Tools:** When you activate any Water Tool, the Water Tools menu appears to provide you with additional functionality.
- **Fence Tools:** When you activate the Fence Tool, the Fence Tools menu appears to provide you with additional functionality.
- **Building Tools:** When you activate the Building Tool, the Building Tools menu appears to provide you with additional functionality.

Tools Palette

The Tools Palette gives you quick access to all the key editing tools in CityScape. Each icon on the Palette represents one or more closely related tools. For example, Geography Tools include the Geography Circle Tool, Geography Polygon Tool, and Geography Grid Tool among others.

To active a tool, LEFT-CLICK the icon of the tool in the Tools Palette, which appears on the left of the 3D Render Window by default. In the case of tool groups, LEFT-CLICK the drop-down menu next to the icon, and select the desired tool from the list.

3D Render Window

The 3D Render Window displays the currently loaded terrain via a simulated camera that you can move around, as described in “Using the Camera” later in this chapter.

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Main Widget Panel

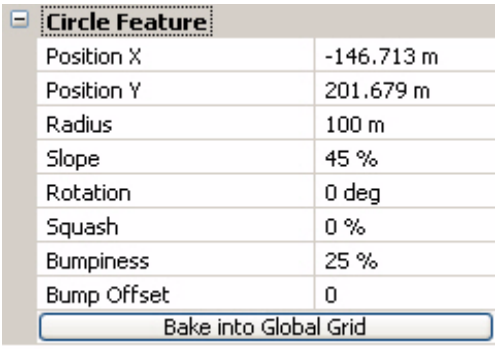
The Main Widgets give you quick access to functionality that complements the active tool. The Main Widgets change whenever you switch active tools or select different types of objects.

For the sake of simplicity, this Quick Start does not document the functionality of each and every individual widget. Later chapters on using specific tools document, in detail, every widgets for the tool of interest. However, going through this tour will teach you how to use enough widgets to accomplish key tasks. To get familiar with the various ways you can change the values in widgets, see Using the Widgets below.

Using the Widgets

You can change the values of the widgets in different ways, depending on the type of widgets. Regardless of the type and method of adjustment, the final value will always be clamped between predefined min and max values.

- To increase or decrease the value of a widget by a single step, LEFT-CLICK the UP/DOWN BUTTONS that appear to the right of the widget after you click in its value field.
- To type in an exact value, LEFT-CLICK the CURRENT VALUE in the widget, and then TYPE the NEW VALUE.
- To smoothly scroll the value of a specific widget, LEFT-CLICK-HOLD the TITLE of that widget and DRAG the MOUSE:
  - For example, to scroll the value of the “Slope” widget, LEFT-CLICK-HOLD on the word “Slope” in the Widget Panel (see figure below), and DRAG the MOUSE left or right. While you drag, the cursor disappears so it won’t hit the edge of the screen or get lost.



  - When scrolling, the value changes by a step size that is appropriate to that widget. For example, if the range of a value is from 0 to 1, the step size is typically fixed at 0.01. For a widget with a range -100 to +100, the step size would be a lot larger.
    - » HOLD the SHIFT-KEY while scrolling to increase the step size by 10x the default step.
    - » HOLD the CTRL-KEY while scrolling to decrease the step size to 1/10th the default.

Library Browser

Some tools involve the use of external items, such as the Prop Tools (i.e. you're working with imported 3D models with textures). These tools provide you with a version of the Library Browser. You use the Library Browser to import new items, modify their global properties, manage them, and select specific items to place into terrains.

The Library Browser takes slightly different forms with specific tools: With Road Tools, the Library Browser becomes the Road Style Browser and gives you access to road styles. With the Material Tool, the Library Browser becomes the Material Browser and gives you access to materials that can use imported textures. With the Prop Tool, the Library Browser becomes the Prop Browser and gives you access to imported 3D props. A few other tools also use a version of the Library Browser, while some tools do not use it at all.

For background information on the Library Browser, and the items it provides access to, see "Chapter 4: The Project System" in the *CityScape User Manual*. For details on using the library browser with a specific tool, see the chapter dedicated to that tool. (For example, for details on using the Road Style Browser with the Road Tools, see Chapter 6, section 2, "Using the Road Style Browser" in the CityScape User *Manual.)*

Using the Camera

The camera system is designed for you to quickly view the contents of terrains from any perspective. You must HOLD the ALT-KEY whenever you move the cameras. Press to toggle between Perspective and Orthographic Cameras:

- The Perspective Camera uses a polar model in which the camera orbits around the point of interest marked by a faint plus (+) in the center of the Render Window.
- The Orthographic Camera looks straight down on the terrain at all times (i.e. helicopter view). All sense of perspective is removed, so you have an orthogonal (flat) view down on everything that's visible in the viewport.

Perspective Camera

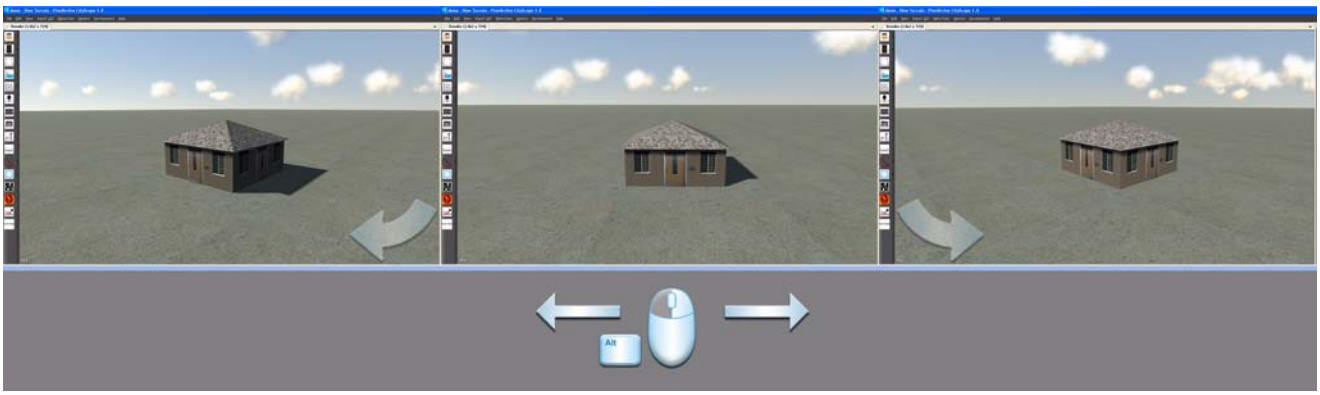
As a type of polar camera, the Perspective Camera is described by its azimuth angle, inclination angle, and distance from the point of interest. You can switch to the Perspective Camera by selecting **View > Cameras > Perspective Camera**. You can toggle between the Perspective and Orthographic cameras with by pressing the **C-KEY**.

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Rotate the Perspective Camera

When rotating the camera, the point of interest remains fixed. The camera orbits the point of interest without changing the distance between them.

1. To rotate the camera around the current point of interest, HOLD both the ALT-KEY and LEFT-MOUSE BUTTON, and then DRAG the MOUSE.



Zoom the Perspective Camera

When zooming, the camera moves closer to, or further from, the point of interest. The point of interest remains fixed.

1. To zoom the camera, HOLD both the ALT-KEY and RIGHT-MOUSE BUTTON, and then DRAG the MOUSE. If you have a mouse with a scroll wheel, you can also HOLD the ALT-KEY and rotate the wheel to zoom.



Note: *When you "zoom" a lens on a traditional camera, the magnification changes but the distance between the camera and point of interest do not change.*

In contrast, when you “zoom” the camera in CityScape, you are actually moving the camera itself closer to, or further from, the point of interest. We use the term "zoom" anyway because it’s widely understood, but a more precise technical term would be the verb form of “dolly,” as it’s used in the motion picture industry.

The distance between the camera and point of interest affects your perspective on the terrain. Helpful hints:

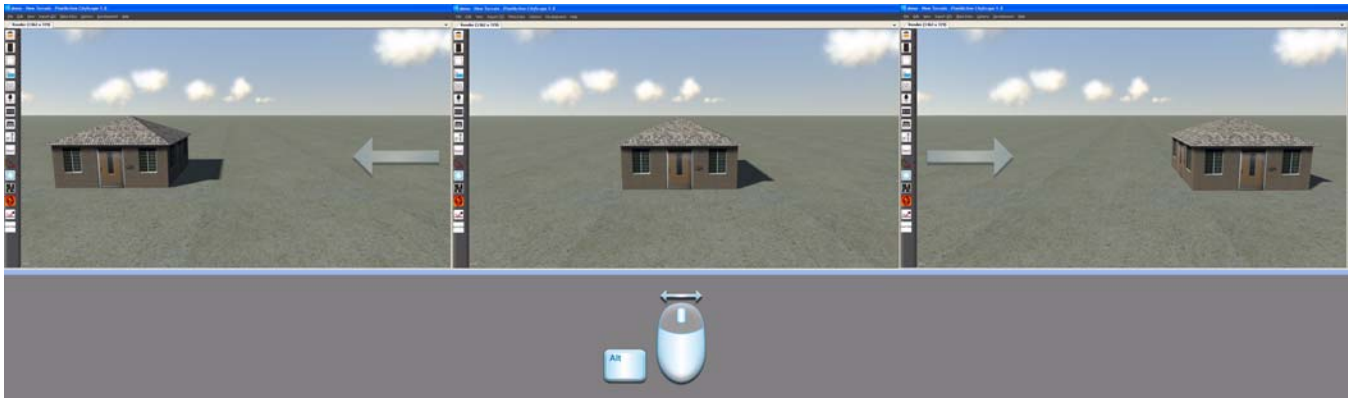
- To view from the point of view of a person standing on the terrain, zoom the camera all the way in until the camera stops. At this point, the position of the camera will be fixed approximately 2 meters above the terrain, so that rotating the camera will have the effect of turning your head while standing on the terrain.

- To view the terrain from an overhead perspective, try zooming the camera further back first, or switch to the Orthographic Camera.

Translate the Perspective Camera

Translating the camera moves the camera along with the point of interest, while preserving the distance between the two. The translation is limited to a plane that's perpendicular to the direction vector of the camera.

1. To translate the camera, HOLD the ALT-KEY, the LEFT-MOUSE BUTTON, and the RIGHT-MOUSE BUTTON at the same time. Then DRAG the MOUSE in any direction.



After you translate the camera, rotating and zooming the camera will operate relative to the new point of interest.

> ***Tip:*** *Translate the camera until the "+" in the middle of the Render View centers on the desired point of interest the terrain. Then zoom and rotate the camera to the desired perspective.*

ORTHOGRAPHIC CAMERA

The Orthographic Camera looks straight down on the terrain. All sense of perspective is removed, so you have an orthogonal (flat) view down on everything visible in the 3D Render Window. North is always towards the top of the screen.

You can switch to the Orthographic Camera by selecting **View / Orthographic Camera** from the Main Menu. You can also press the **C-KEY** to toggle between Perspective and Orthographic Cameras.

Zoom the Orthographic Camera

1. To zoom the Orthographic Camera, HOLD the ALT-KEY and RIGHT-MOUSE BUTTON, and then DRAG the MOUSE.

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Translate the Orthographic Camera

Since the look-at vector of the Orthographic Camera is always perpendicular to the ground, translating this camera moves it in a plane parallel to the ground. Put another way, translating the Orthographic Camera maintains the same elevation, like a helicopter flying level.

1. To translate the camera, HOLD the ALT-KEY and LEFT-MOUSE BUTTON, and then DRAG the MOUSE. The point of interest moves along with the camera.

CREATING AND EDITING WORLDS

You do most of the work in CityScape with editing tools. Examples:

- Use the Geography Tool to create and modify hills and valleys by combining circles, polygons, grids, and other geometric primitives. When you export the terrain, CityScape tessellates all geometric primitives into triangles.
- Use the Road Tool to create and modify roads in the terrain.
- CityScape includes many other tools for creating robust 3D worlds. This Quick Start covers the basics of using the most common tools:

- Using Geography Tools, next
- “Using Road Tools” on page 19
- “Using Water Tools” on page 33
- “Using Material Tools” on page 35
- “Using Prop Tools” on page 37
- “Using Sky Tools” on page 45
- “Using GIS Tools” on page 46

Initially, the Geography Tool is active by default. The name of the active tool is displayed at the bottom left corner of the 3D Render Window.

To activate a tool, LEFT-CLICK its icon in the Tools Palette on the left of your screen, by default. You can change tools at any time.

When you switch tools, the Main Widgets change to display the specific features and options that are available with the current tool. The Main Widgets also change when you select a different type of object in a terrain.

Using Geography Tools

When you initially run CityScape, the Geography Tools will be active by default. You can use Geography Tools to shape the underlying terrain. To accomplish this goal, you place and customize Circle Features, Polygon Features, and Grid Features that CityScape converts to 3D hills and valleys. You can also "paint" hills and valleys directly into the terrain, and use the Main Widgets to customize all geography features.

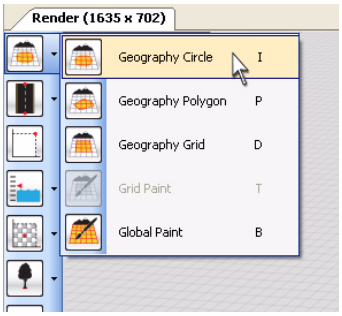
The Geography Circle Tool

A Circle Feature is an ellipse that causes the terrain within it to rise or lower to form a hill or valley. A Circle Feature starts life as a circle, but you can make it into an ellipse by changing the *Squash* parameter in the Main Widgets for Circle Features.

The Geography Circle Tool lets you place and customize Circle Features in the terrain. Like all geography features, the overlapping Circle Features are additive.

Create Circle Features

To place a Circle Feature in the terrain, follow these steps:

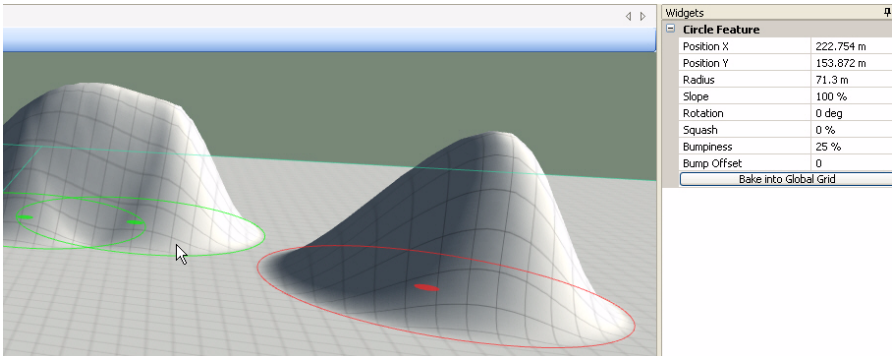


1. With a terrain loaded in CityScape, activate the **Geography Circle Tool** by CLICKING its ICON in the Tools Palette.
2. To place a Circle Feature in the terrain, RIGHT-CLICK at the target location in the terrain.
3. Try placing a few Circle Features – some separate, some overlapping.

Select Circle Features

Editing Circle Features requires that you first select one or more of them with the Geography Circle Tool:

- To select one existing Circle Feature, LEFT-CLICK its **center point**. It will change from green (unselected) to red (currently selected). In the figure below, the red Circle Feature in the Render Window has been selected, and the Circle Feature Widgets appear on the far right.



- To select several existing Circle Features at the same time, LEFT-CLICK-HOLD and DRAG THE MOUSE to draw a box around their center points.

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Edit Circle Features

After you create a geographic feature, you can always edit it later (unless you intentionally bake it into the terrain). To edit a Circle Feature, first select it and then proceed as follows:

- To move a selected Circle Feature, hover the mouse arrow over its center point, LEFT-CLICK-HOLD, and DRAG THE MOUSE.
- To delete a Circle Feature, select its center point and press the DELETE-KEY.
- Select some Circle Features and experiment with modifying the values of the Circle Feature Widgets. Examples:

  Slope Widget: *Controls the height and steepness of Circle Features. Big positive slopes make high hills, and negative slopes make valleys (that can optionally be filled with water later). Setting the Slope widget to 100% for hills, or -100% for valleys, makes the Circle Feature as high or low as it can be by itself. To go higher or lower, create multiple overlapping Circle Features.*

  Squash Widget: *Controls the eccentricity of the Circle Feature. If Squash is non-zero, the circle becomes an ellipse.*

THE GEOGRAPHY POLYGON TOOL

The Geography Polygon Tool lets you place and customize Polygon Features in the terrain. A Polygon Feature is a polygonal area defined by 3 or more points connected by line segments. You can use Polygon Features to make natural looking mountains and valleys with irregular contours.

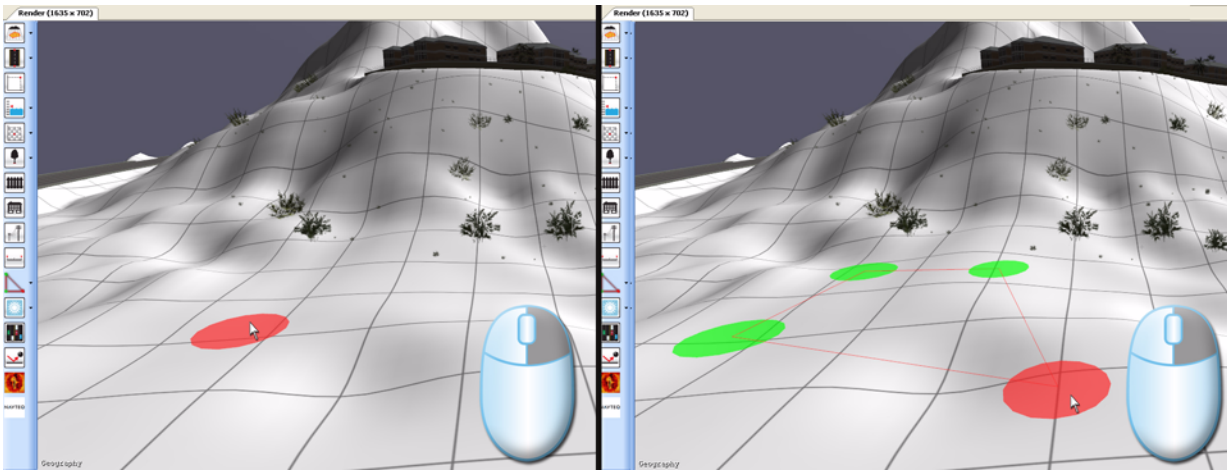
Create Polygon Features



Follow these steps to place a Polygon Feature in a terrain:

1. With a terrain loaded in CityScape, activate the **Geography Polygon Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)
2. RIGHT-CLICK at the location in the terrain where you want to place the first point in the polygon.

3. Continue RIGHT-CLICKING a few times to place new points and expand the polygon.

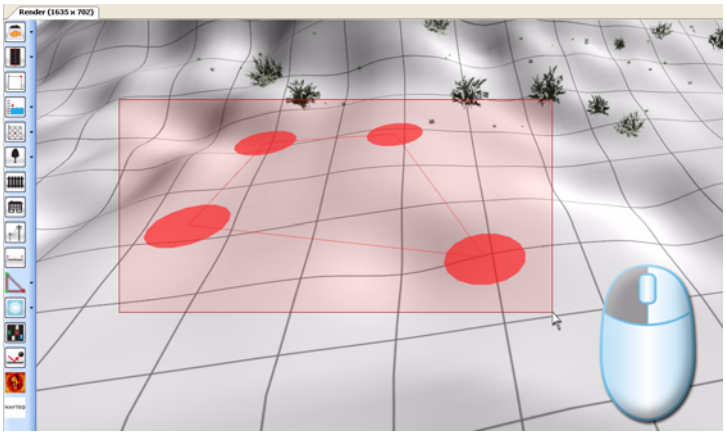


4. To start a new polygon, first select nothing by LEFT-CLICKING away from selectable objects in the terrain. Then repeat the steps above.

Select Polygon Features

Editing a polygon requires that you first select one or more points with the Geography Polygon Tool:

- To select one point in a Polygon Feature, LEFT-CLICK the point. The figure in the "Edit Polygon Features" section shows one actively selected point (red) and the other points that are not selected (green).
- To select multiple points (first method), HOLD the SHIFT-KEY and LEFT-CLICK each point in succession.
- To select multiple points (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to draw a pick box around them.

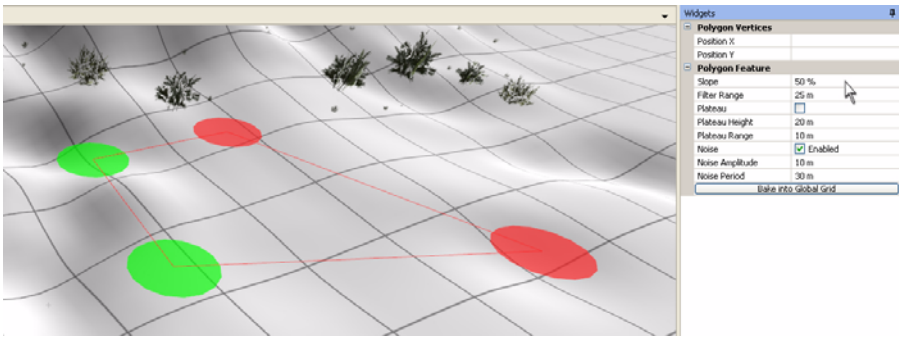


- To select an entire polygon, DOUBLE-LEFT-CLICK any point in it.

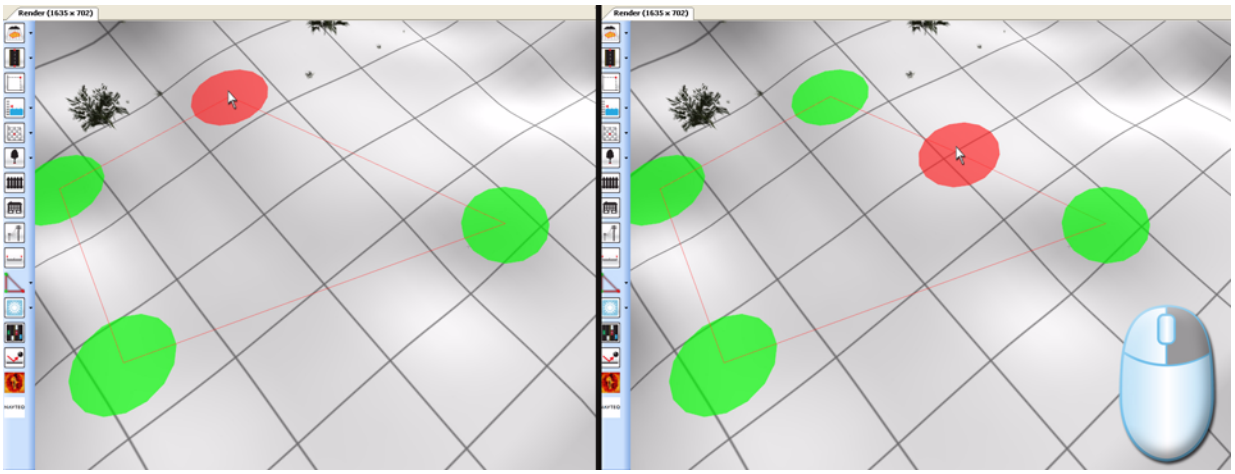
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Edit Polygon Features

As with all geographic features, you can edit the properties of Polygon Features by selecting one or more points in an existing polygon, and then using the Main Widgets to modify the parameters controlling the polygon.



- To delete one point on an existing polygon, select the point and then press the DELETE-KEY.
- To move a point on an existing polygon, be sure the mouse is hovering over the selected point, and then LEFT-CLICK-HOLD the point and DRAG THE MOUSE.
- To move multiple points, first select the points as described in “Select Polygon Features”. Hover the mouse over one of the selected points, and then LEFT-CLICK-HOLD and DRAG THE MOUSE.
- To add a new point in an existing polygon, first select the appropriate node (i.e. the new node will be created clockwise relative to the selected node.) Then RIGHT-CLICK to add the new point to the polygon.



- Select a Polygon Feature and experiment with modifying the parameters of the Polygon Feature Widgets. Most of the parameters affect the entire polygon, even if you select only one point in it.
- To select the entire polygon, DOUBLE-CLICK any node in the polygon.

THE GEOGRAPHY GRID TOOL

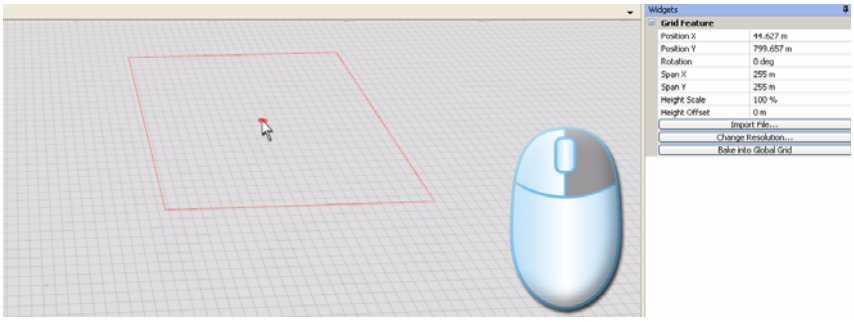
The Geography Grid Tool lets you place and customize a Grid Feature in the the terrain. A Grid Feature is a rectangular area controlled from its central point. You can raise and lower the terrain within a Grid Feature to form hills or valleys. Grid Features are also useful for importing localized height-maps based on 2D images, and hand-painting 3D terrain, as explained below.

Create Grid Features

To create a Grid Feature, follow these steps:

1. With a terrain loaded in CityScape, activate the **Geography Grid Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)

2. To place a new Grid Feature, RIGHT-CLICK the location in the terrain where you want the grid to be centered. The figure below shows a Grid Feature actively selected in the terrain along with its Main Widgets.



Select and Edit Grid Features

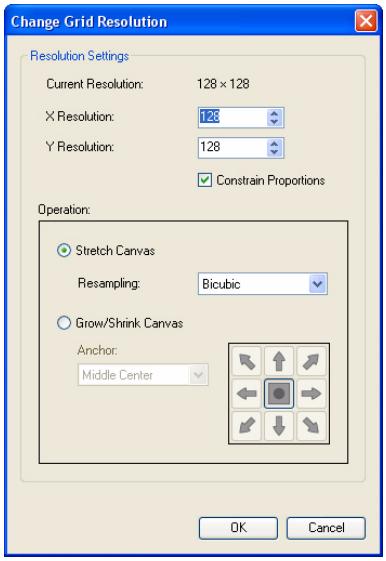
Use the Geography Grid Tool to select and edit a Grid Feature:

1. To select an existing Grid Feature, LEFT-CLICK its **center point**.

2. To move a selected Grid Feature, be sure the mouse arrow is hovering over its center point, and then LEFT-CLICK-HOLD and DRAG THE MOUSE.

3. To delete a Grid Feature, select it and press the DELETE-KEY.

4. When a Grid Feature is selected, you can change its properties using the Grid Feature Widgets. Examples:
   – To adjust the size of the selected grid, change the **Span X** and **Span Y** widgets.
   – To modify the resolution of the currently selected Grid Feature, LEFT-CLICK the **Change Resolution** button in the Widgets to display the Change Grid Resolution dialog box. In the dialog, change the values of the **X-Resolution** and **Y-Resolution** fields in the Change Grid Resolution dialog. (Note: Changing the resolution of a Grid Feature does not change its size.)



Import Non-GIS Height Maps

CityScape can convert almost any 2D image to a height field (i.e. hills and valleys) via the Geography Grid Tool. (For details on geo-located height maps, see “Chapter 21: Importing GIS Data” in the CityScape User *Manual.*) When you import an image, CityScape first converts it to grayscale. Then CityScape converts black to ground level, white to the highest elevation allowed by the **Height Scale** widget, and grays to intermediate elevations according to their brightness.

CityScape supports JPEG, TIFF, and most other common 2D image formats. To import an ordinary 2D image as a height map, follow these steps:

1. Activate the **Geography Grid Tool**, if necessary.

2. LEFT-CLICK the **center point** of an existing Grid Feature to select it. Or create a new Grid Feature.

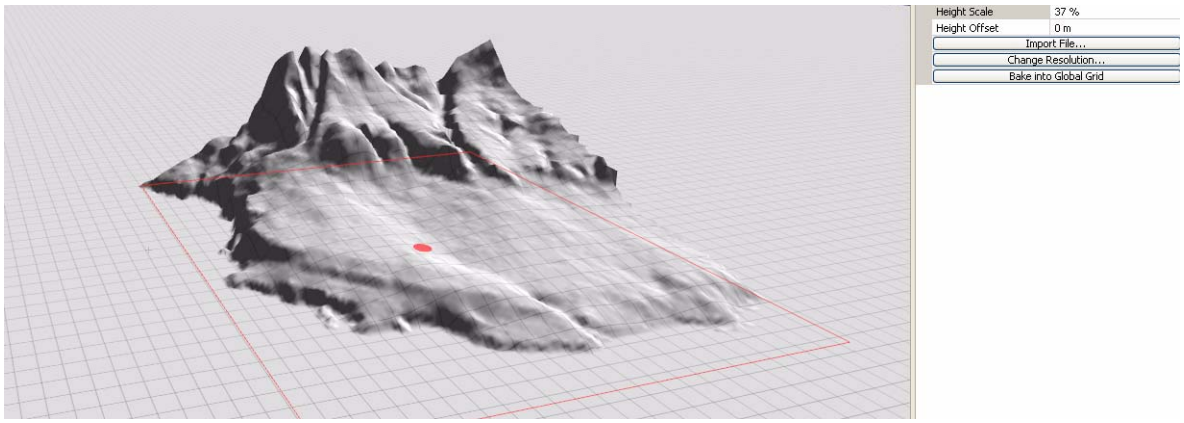
3. Press the **Import File** button in the Main Widgets for the Grid Feature. In the file browser that appears, locate and select the **SampleHeightmap.jpg** image in the following folder, by default:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds\heightmaps

or here for WinXP users:

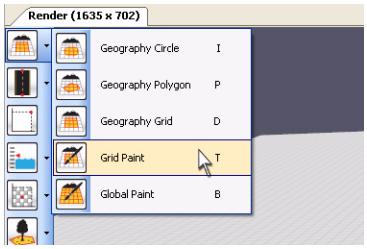
C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds\heightmaps

4. CityScape converts your image into 3D geography within the selected Grid Feature.



THE GEOGRAPHY GRID PAINT TOOL

To "paint" changes to 3D terrain within a Grid Feature, follow these steps:

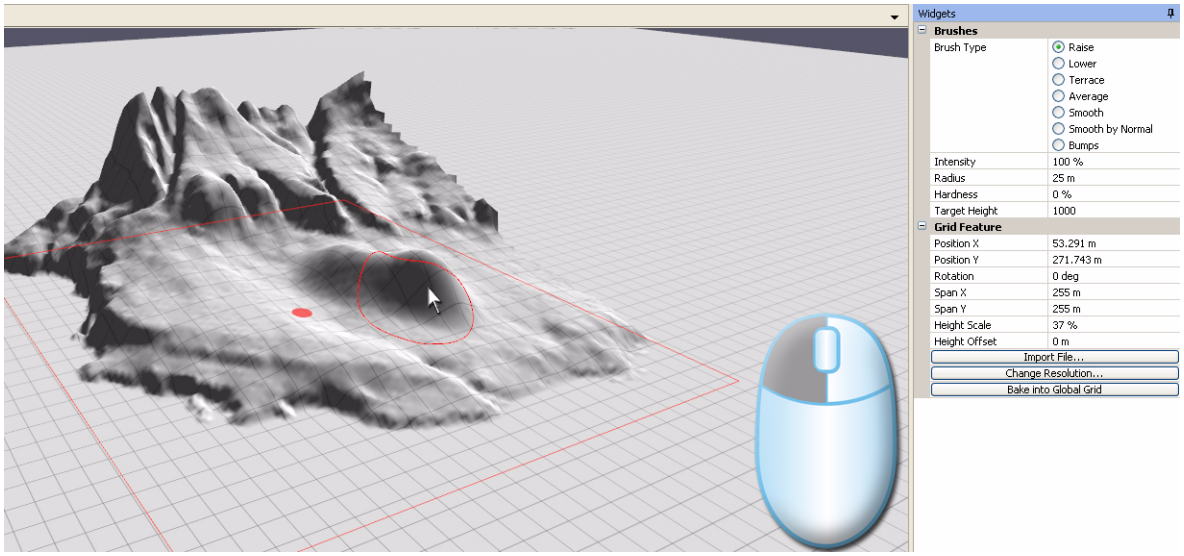


1. In CityScape, open a terrain with at least one Grid Feature, or create a new Grid Feature in the open terrain.

2. Select a Grid Feature in the terrain.

3. Activate the **Grid Paint Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)

4. To paint 3D terrain on the selected Grid Feature, LEFT-CLICK-HOLD and DRAG the MOUSE within the Grid Feature. The resulting 3D terrain is part of the selected grid, and can be moved with the grid and edited with the Main Widgets for the Grid Feature, as shown below.



5. Experiment with modifying the values of the Main Widgets. Try changing brush types and adjusting their sub-parameters. The figure above shows the results of 3D painting with the Raise brush. Try painting with various brushes.

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The Geography Global Paint Tool

You can "paint" changes to 3D terrain anywhere in the open terrain in much the same way as painting within a Grid Feature. Here's how:

1. With a terrain loaded in CityScape, activate the **Global Paint Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the drop-down menu next to the main Geography icon.)

2. To paint changes to 3D terrain, LEFT-CLICK-HOLD and DRAG the MOUSE anywhere on the terrain. The resulting 3D terrain is part of the global grid, so this terrain cannot be moved or edited with widgets after you paint it.

3. Experiment with modifying the values of the Global Paint Widgets. Your changes will affect future painting only.

Using Road Tools

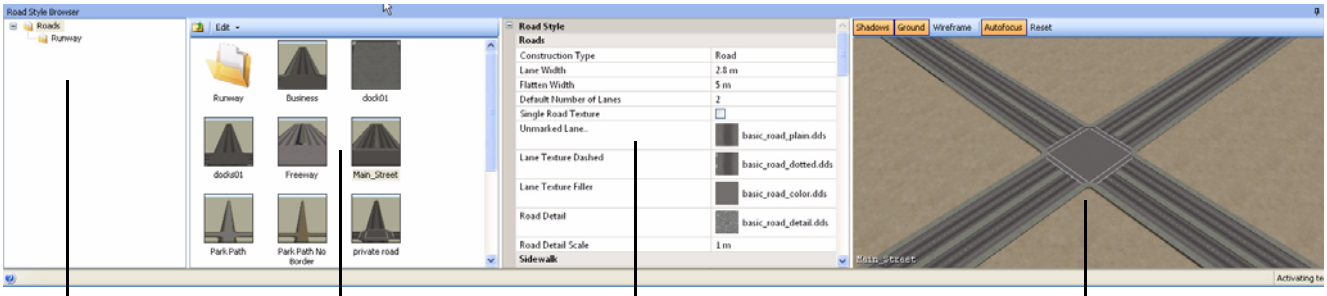
The Road Tools let you create and modify a network of roads and intersections. Some of these tools are in the Road Widgets; some are in the Road Style Browser or Road Tools Menu, which appear when any Road Tool is active.

This Quick Start introduces you to The Road Tool. Other tools in the Road Tools Group include the *Road Tangent Tool* and *Road Height Tool*, both of which are documented in "Chapter 6: Using Road Tools" in the CityScape User Man*ual.*

About The Road Style Browser

Whenever you select any of the Road Tools from the Tools Palette, the Road Style Browser appears in the lower area of the CityScape window, by default.

The Road Style Browser shows the contents of the Road Style Library, so you can modify and place specific road styles in the terrain. You can use road styles to visually differentiate between main streets, freeways, alleys, etc. You can also import new textures for road styles, and otherwise edit road styles.



Road Style Tree Road Style Icons Road Style Sub-Widgets Road Style Preview

To select a road style, simply LEFT-CLICK the ICON of the desired style in the Road Style Browser. With the Demo Project, try selecting the **Main_Street** road style. A sample of the style is previewed in the right sub-window.

About Road Style Sub-Widgets

The Road Style Sub-Widgets are part of the Road Style Browser at the bottom of the screen, by default. These Sub-Widgets allow you to edit the *global* properties of a road style, regardless of whether or not that road style is currently used in the open terrain.

The Prop Browser and Material Browser also have sub-widgets for editing the global properties of props and materials, respectively. In all cases, Sub-Widgets control the global properties of an object. Hence, changing these values will affect all instances of the object in all terrains sharing the same project file.

In contrast, the Main Widgets in the upper right of your screen, by default, let you edit only the *local* properties of objects that are actively selected in the open terrain.

For example, in the figure above, *Default Number of Lanes = 2* means that every new road using the selected **Main_Street** road style will begin life with 2 lanes in each direction. After you create some roads in a terrain, you can locally customize the number of lanes of each road using the Main Widgets.

THE ROAD TOOL

To use the Road Tool, you must understand road nodes and road segments: By definition, a “road node” is a point that defines the start or end of a single segment of road. An entire road consists of a series of nodes attached by segments. An intersection can optionally exist at a road node.

The usage of the Road Tool is organized into the following sections, which we recommend you read in order:

- Creating Your First Road
- Creating Additional Roads
- Continuing an Existing Road
- Smoothing Out a Road
- Close a Road Loop
- Select and Edit Road Nodes
- Select and Edit Road Segments
- Create a 2-way Intersection
- Create a 3-way Intersection
- Create a 4-way Intersection
- Link and Unlink Segments Across Intersections
- Create Merges and Splits
- Build Bridges and Overpasses

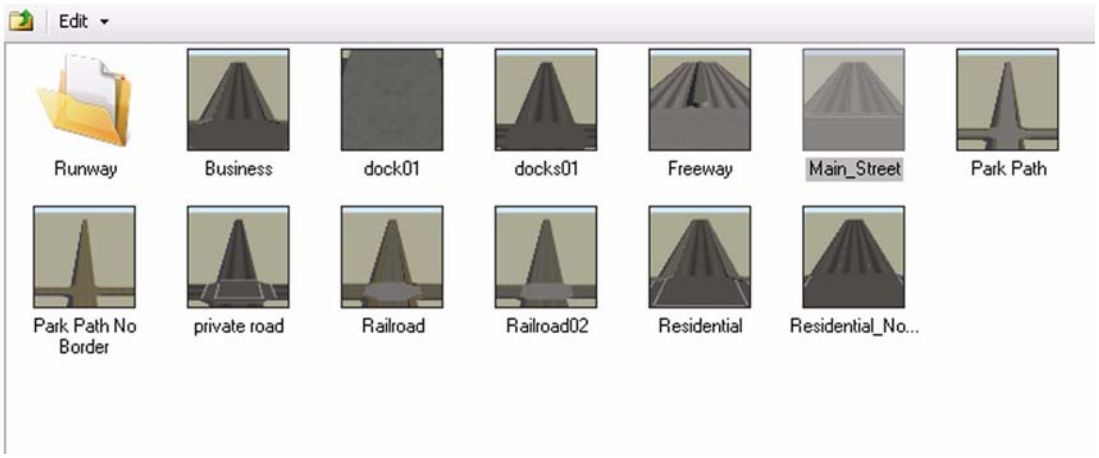


Creating Your First Road

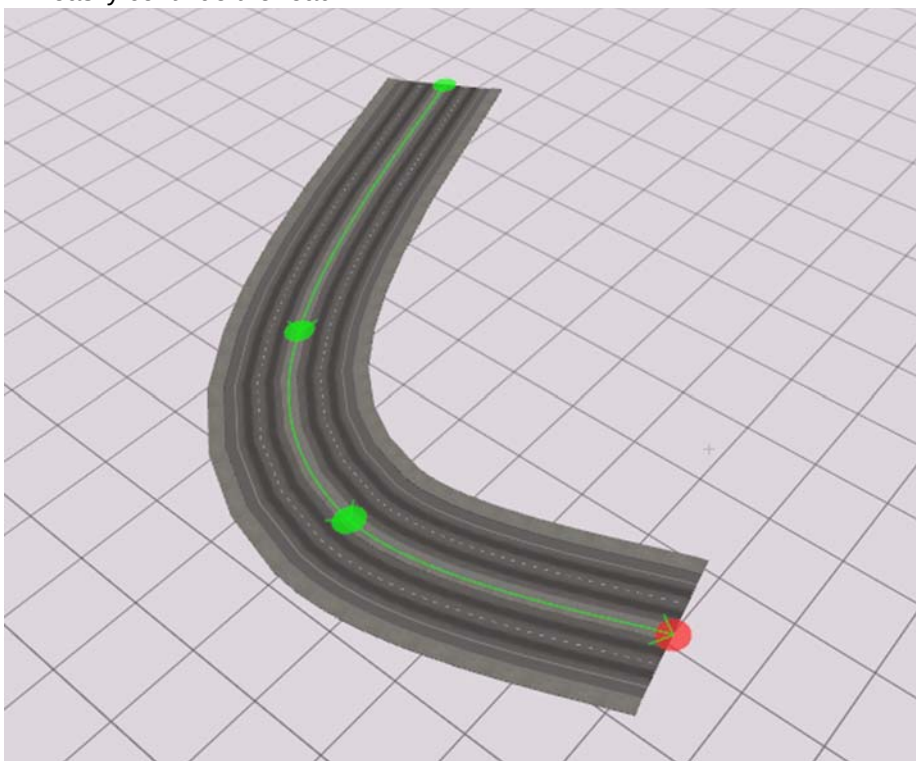
To create a road, proceed as follows:

1. Activate the **Road Tool** by CLICKING its ICON in the Tools Palette.

2. In the Road Style Browser, which appears when any of the Road Tools are active, select a style in the Road Styles Icons (the example icons below may not correspond to your road styles).



3. RIGHT-CLICK a few times on the terrain to lay down a few road nodes. Each time you click in an open area to place a new road node, CityScape adds a road segment connecting the previous node to the new node. The last road node you placed is always the current selection, so you can easily continue the road.



Creating Additional Roads

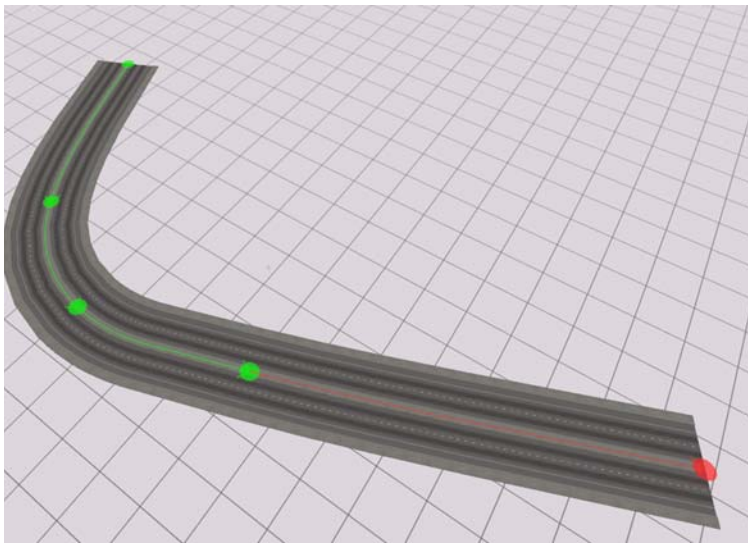
To start a new road that is not connected to the last road, proceed as follows:

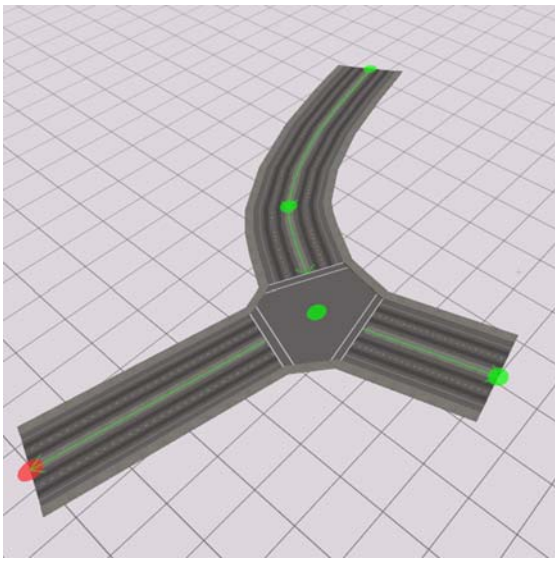
1. First LEFT-CLICK in an area without objects to deselect everything.
2. Then LEFT-CLICK the desired road style in the Road Style Browser to select it.
3. RIGHT-CLICK a few times to place new road nodes with interconnecting segments.

Continuing an Existing Road

If you deselect a road, and later want to resume work on that road, follow these steps:

1. To select the road node from which to resume, LEFT-CLICK the **target node** to select it.
   – If you resume from an end node, then placing new nodes will continue the same road, like the left image below. If you resume from any node other than an end node, then placing new nodes will cause a new road to branch out from an intersection at the selected node, like the right image below.





2. RIGHT-CLICK a few times to place some new nodes as usual.

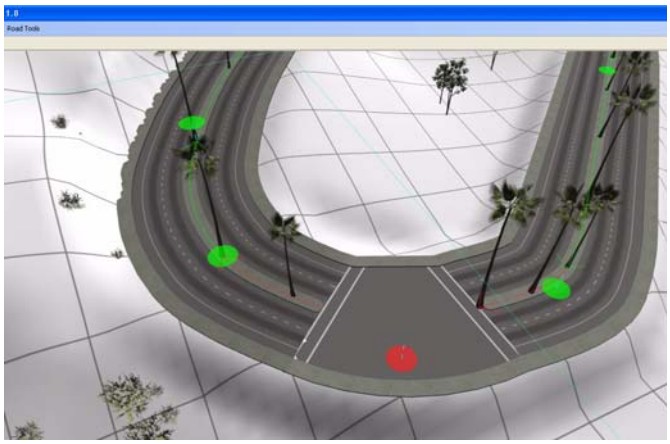
Smoothing Out a Road

If a 2-way intersection appears in your road (like the left image below), but your desired goal is to have a smooth road without an intersection (like the right image below), follow these steps:

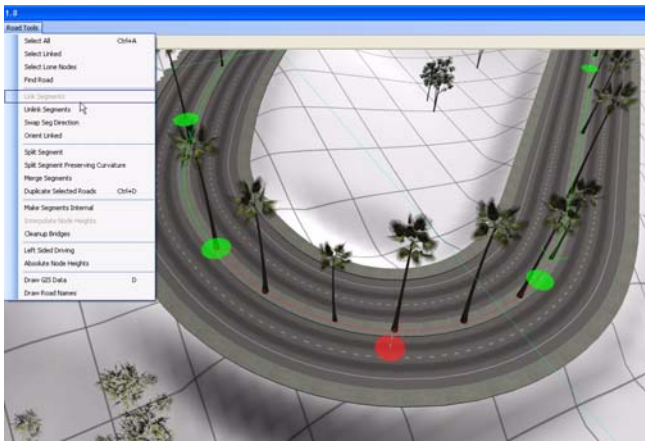
1. LEFT-CLICK-HOLD and DRAG the MOUSE to draw a pick box around the node in the intersection, as well as the segments on either side of the intersection, to select them all.

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2. Select **Link Segments** from the Road Tools Menu in the Main Menu. (The Road Tools Menu appears only when a Road Tool is active.) The 2-way intersection will be replaced with a continuous road.



2-way intersection before linking segments on either side of the selected (red) node.

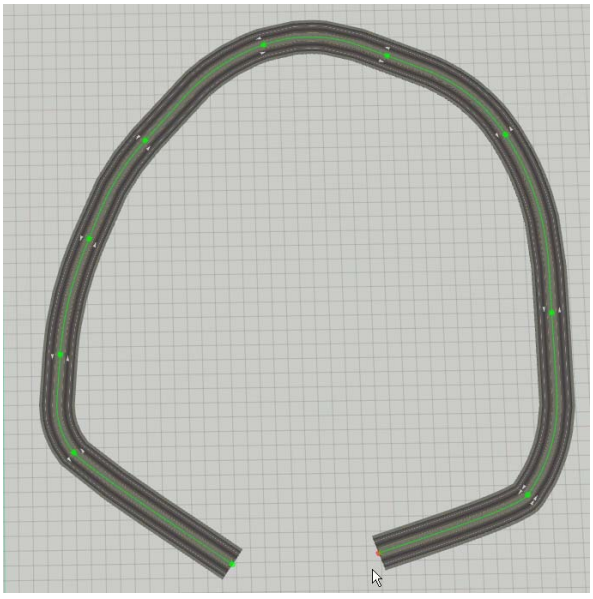


After linking segments, the 2-way intersection is replaced by smooth road through the selected node.

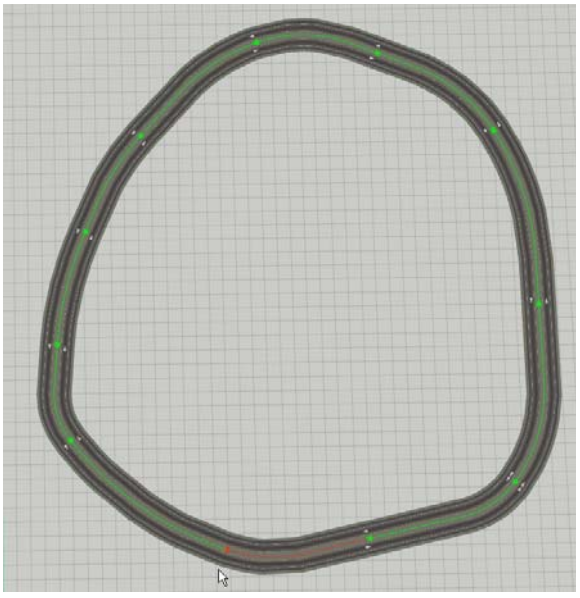
Close a Road Loop

Some features of CityScape call for a closed loop of road. There are many ways to create closed loops of road, but the simplest case is a single oval road without intersections:

1. Create an oval road, such as the one shown in the right figure below.



An open loop road with the end node selected.



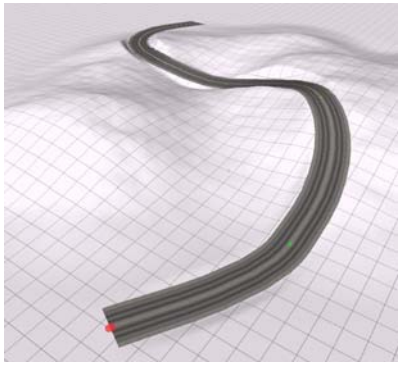
After linking the end node to the start node, the road becomes a closed loop.

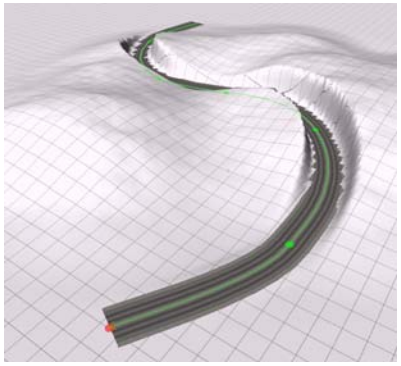
2. Select the end node. Then RIGHT-CLICK directly on the node that started the road. This action connects the end node to the start node, closing the loop of road. The right figure above shows the closed loop.

Select and Edit Road Nodes

Some properties of roads are specific to road nodes, and other properties are specific to segments. This topic touches on some useful properties of nodes:

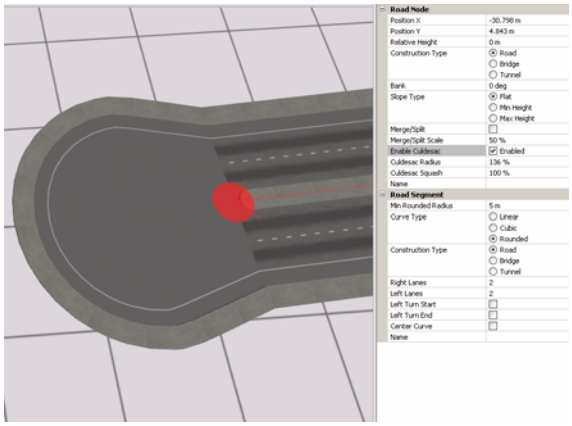
- To select one road node, LEFT-CLICK the target **node** (it will turn red).
- To select multiple nodes (first method), HOLD the SHIFT-KEY and LEFT-CLICK each node in succession.
- To select multiple nodes (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to enclose the target nodes in a pick box. This method will also select the segments within the pick box.
- To move a section of road, first select the node(s) in the section of road you wish to move. Then hover the mouse arrow over one of the selected nodes, and LEFT-CLICK-HOLD and DRAG THE MOUSE.
- To delete one or more nodes, select them and PRESS the **DELETE-KEY**. Deleting a node will also delete the segments attached to it. If your road was a closed loop before, deleting a node will open the loop.
- Use the Road Node Widgets to adjust the properties of currently selected nodes. Here are some useful properties that you can adjust with widgets:
  - **Absolute / Relative Height:** Absolute height controls the elevation of the node, relative to the absolute zero elevation. Relative height controls the elevation of the node, relative to the surrounding terrain.
    - » To switch between absolute and relative heights, toggle Absolute Node Heights in the Road Tools Menu of the Main Menu.
    - » The choice of absolute or relative can dramatically affect the way roads pass through mountains and valleys. The examples below show the same road with Relative Height = 0 (left image) and Absolute Height = 0 (right image).



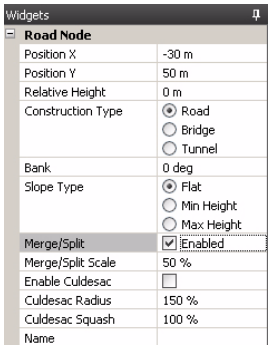


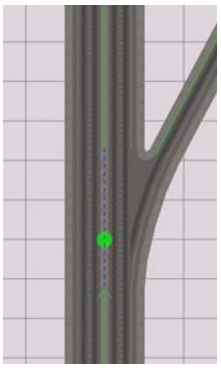
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– **Enable Culdesac**: allows you to convert an end node of an open road into a cul-de-sac.



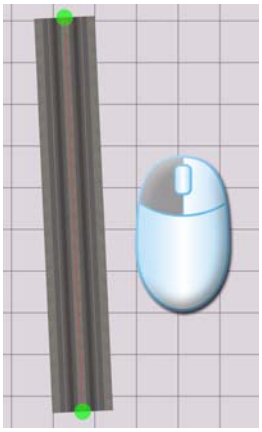
– **Merge / Split:** starts or ends a Y-branch without an intersection.
  » After enabling Merge/Split for node in an existing, placing additional nodes creates a new road stemming from the existing road, but without adding an intersection.
  » Similarly, you can merge one road into another without an intersection by enabling Merge/Split in a node of the target road (i.e. the road you're merging into).





Select and Edit Road Segments

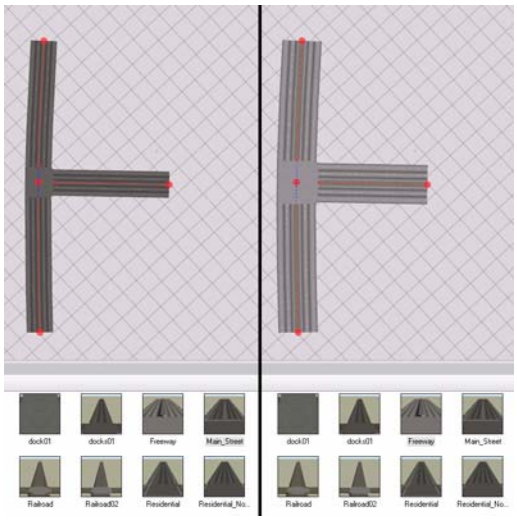
This topic addresses some of the properties of road segments:



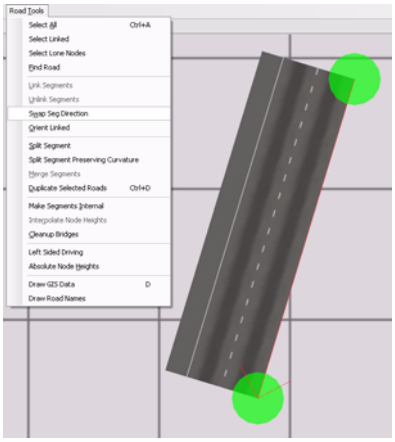
- To select one road segment, LEFT-CLICK the **green line** defining the segment. The line will turn red when it becomes the active selection, as shown in the figure to the left.
- To select multiple segments (first method), HOLD the SHIFT-KEY and LEFT-CLICK **each segment** in succession.
- To select multiple segments (second method), LEFT-CLICK-HOLD and DRAG THE MOUSE to partly or completely enclose the **segments** in a pick box. This method will also select the nodes within the pick box.
- To insert a new node within an existing segment, first select the target segment in an existing road. Then RIGHT-CLICK the **selected segment**.

- To delete one or more segments, select them and PRESS the DELETE-KEY. If your road was a closed loop before, deleting a segment will break the road into an open loop.

- To change the style of all or part of an existing road, select the **road segments** that you wish to change, and then DOUBLE-LEFT-CLICK the new road style in the Road Style Browser.



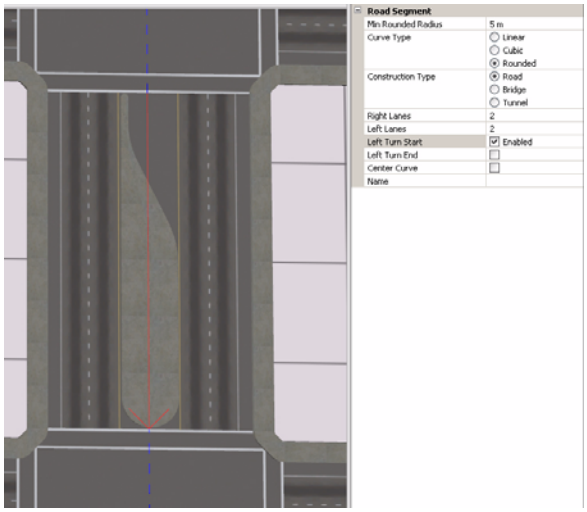
- Every road segment has a direction, which is indicated by an **arrow** at the end of each segment. References to right and left sides of a segment, as well as its start and end, are relative to its direction.
  – To switch the direction of actively selected segments, choose **Swap Seg Direction** in the Road Tools Menu. (The directions of road segments are different than the direction in which traffic will flow.)

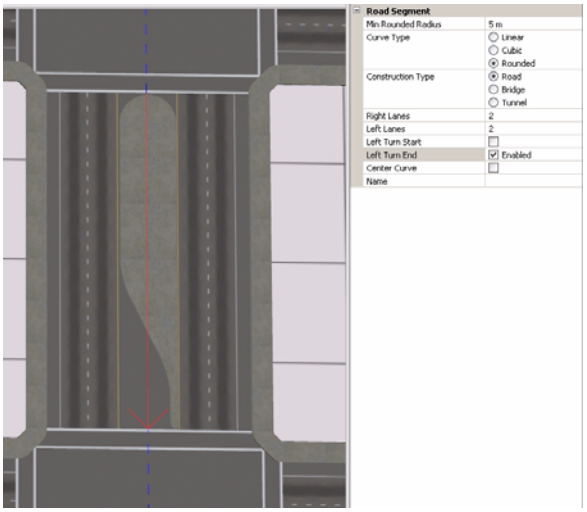


- To switch between right handed driving (default) and left handed driving, toggle **Left Sided Driving** in the Road Tools Menu. (The directions of road segments are different than the direction in which traffic will flow.)

- You can adjust the properties of currently selected segments using the Road Segment Widgets on the right of the screen. Some useful widgets:

- **Right Lanes** determines the number of lanes on the right side of the road, relative to the green arrow indicating the directionality of the road segment. Every road must have at least one (1) right lane.
- **Left Lanes** determines the number of lanes of the left side of the road. For a one-way road, set this widget to zero (0).
- **Left Turn Start** creates a left turn lane in the left side of the road, relative to the segment's direction (see left figure below).
- **Left Turn End** creates a left turn lane in the right side of the road, relative to the segment's direction (see right figure below).





Create a 2-way Intersection

A 2-way intersection might not seem logical, but it can be useful for creating precise 90-degree turns, for example. (Note: Creating a 2-way intersections is the inverse of replacing a 2-way intersection with a continuous road, as explained in "Smoothing Out a Road" on page 22.)

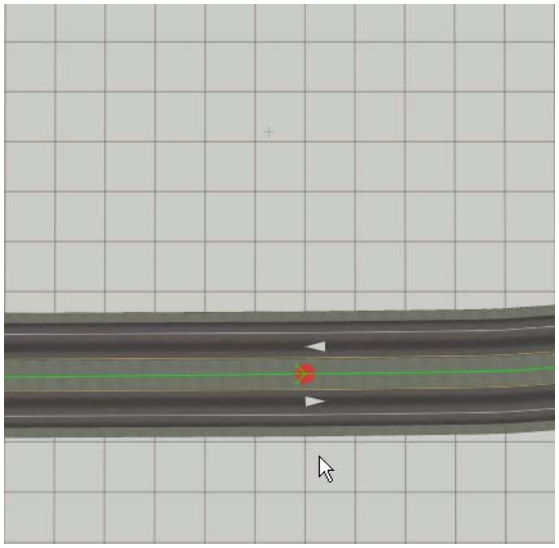
1. In an existing road, select the two segments on either side of the node you wish to change into a 2-way intersection. Selecting the node itself is optional.

2. Choose **Unlink Segments** from the Road Tools Menu in the Main Menu. (The Road Tools menu appears only when a Road Tool is active.) A 2-way intersection is created.

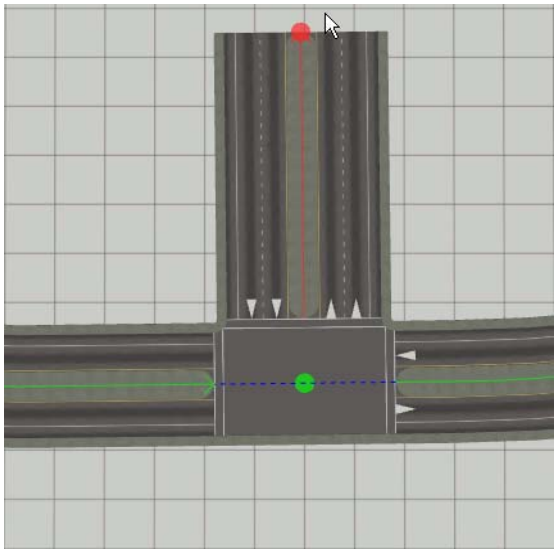
Create a 3-way Intersection

The following procedure is one of many ways to create a 3-way intersection. Such intersections are useful for branching a new road off an existing road, or creating a Y-intersection.

1. In an existing road, select an internal road node (i.e. not an end node) from which you want to branch a new road via a 3-way intersection. Upon being selected, the node turns red.



Steps 1 and 2: Node selected (red), but adjoining segments are not selected.



Step 3: A new road stems from 3-way intersection.

2. Ensure that the segments to either side of the selected node are NOT selected (i.e. green). See the left image in the figure above.

3. LEFT-CLICK again in an open area at roughly right angles to the existing road. A 3-way intersection is automatically created, and a new road branches off the intersection. See the right image in the figure above.

4. Extend the new road as usual.

Create a 4-way Intersection

There are many ways to create 4-way intersections. This section covers two common options.

OPTION 1

1. Create a 3-way T-intersection, as described in Create a 3-way Intersection.

2. Select the road node in the intersection. Make sure that the segments adjacent to this node are not selected.

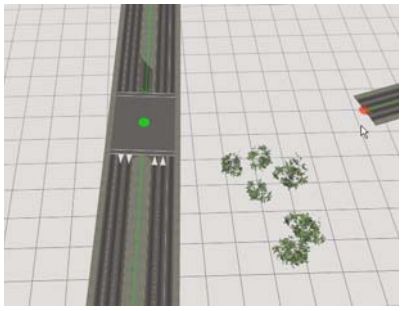
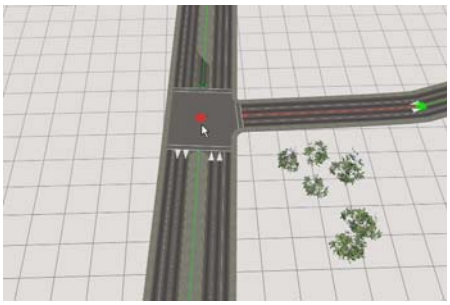
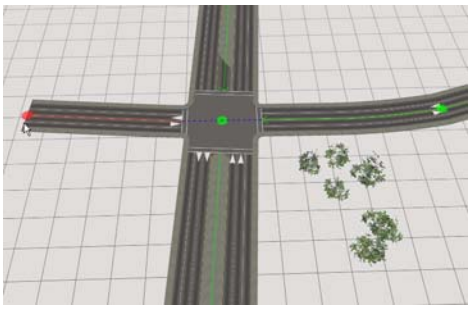
3. RIGHT-CLICK in an open area at roughly right angles to the main road, in the opposite direction as the existing branch, as shown in the figure below. A new road branches off the intersection, and the 3-way intersection becomes a 4-way intersection.

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OPTION 2

1. Ensure that one road already exists (create it if necessary).

2. Create or extend a new road so it crosses the existing road. To form a 4-way intersection at their crossing, extend the new road by LEFT-CLICKING a node in the existing road.

Note: If you do not click a node at the crossing point of the two roads, an intersection will not form. You might not want an intersection in all cases, such as a bridge passing over or under another road.

Link and Unlink Segments Across Intersections

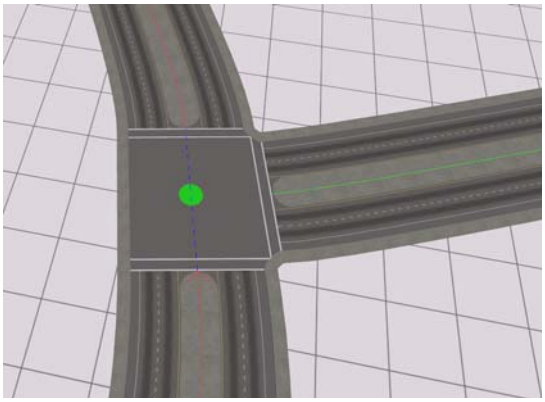
Linking and unlinking segments on either side of an intersection affects the exact positions of incoming and outgoing roads. Also linking could be used by AI software to determine the path of flowing traffic.

When you create any intersection, the segments around it may or may not be linked initially. To manually link or unlink segments across intersections, follow these steps:

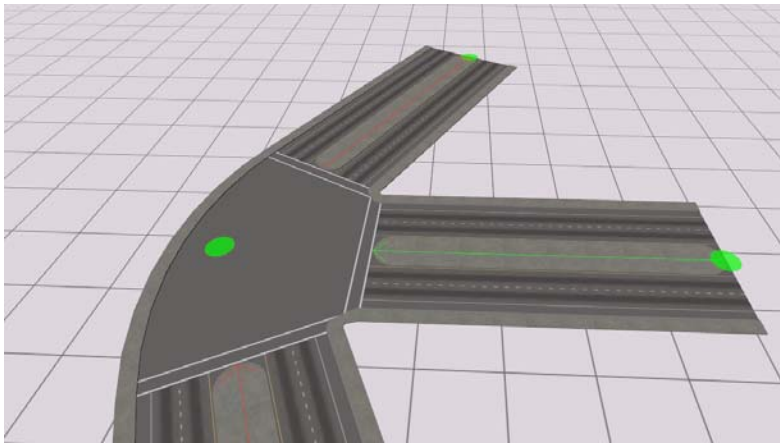
1. Start with an existing intersection, or create an intersection by following the instructions in the previous topics.

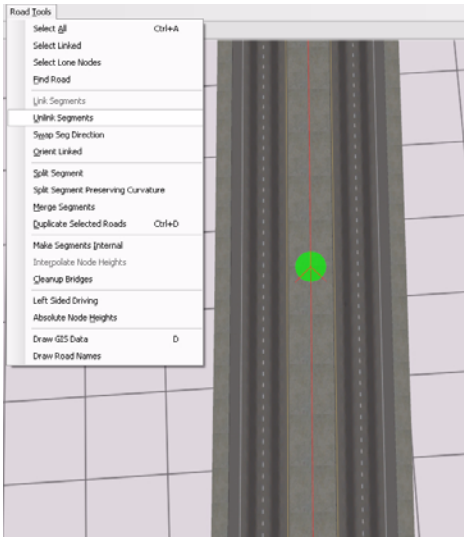
2. Select two road segments that you wish to link or unlink across an intersection. The two segments must share the node at the center of the intersection.

3. To link the two segments, choose **Link Segments** from the Road Tools Menu. A dashed blue line appears through the intersection to show the linking, as illustrated below.



4. To unlink the two segments, choose **Unlink Segments** from the Road Tools Menu. In the illustrations below, the left image shows 3 unlinked segments in a 3-way intersection, and the right image shows 2 unlinked segments in a smooth strip of road.



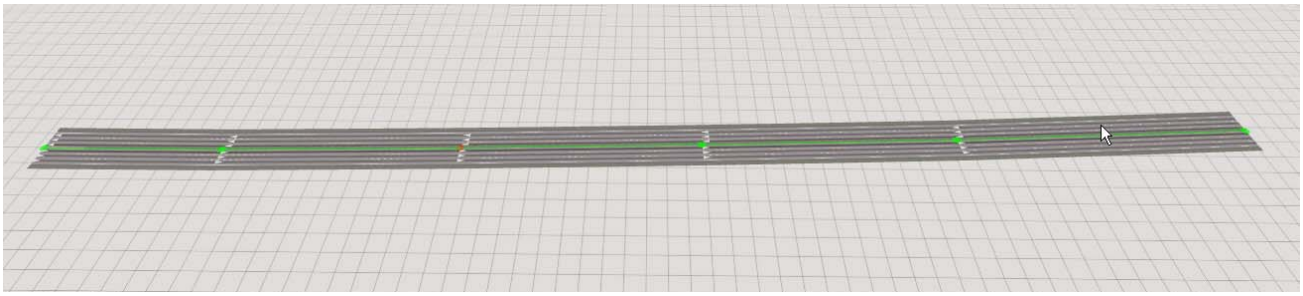


Create Merges and Splits

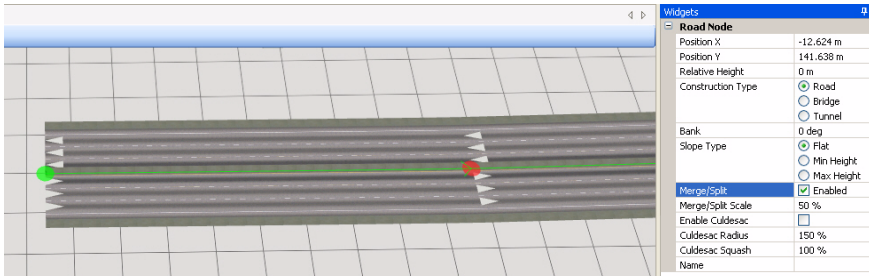
Many types of merges and splits for roads are possible with CityScape. In the exercises below, you will split a ramp off a freeway, and then merge the ramp back into the freeway.

1. In the Road Style Browser, select the **Freeway** road style.

2. In the terrain, lay down six new **road nodes** to form a standalone section of freeway with open ends.

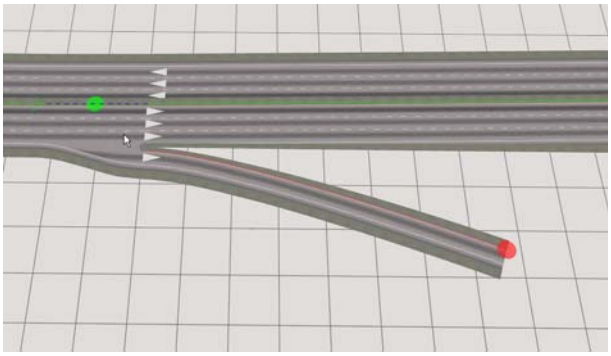


3. Select the **node** located *second from the start*. In the Road Node Widgets, enable **Merge/Split**.

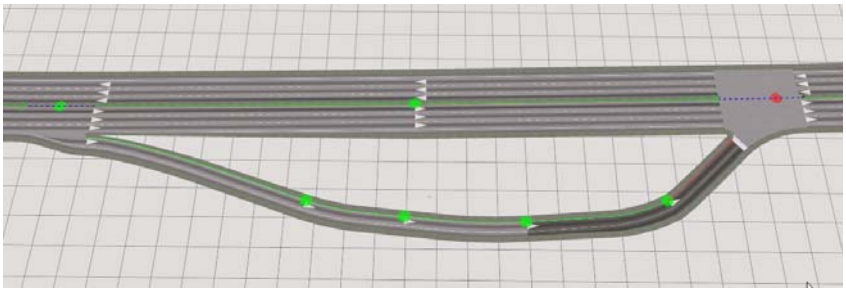


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4. To place a **new node** for the ramp, RIGHT-CLICK to the right of the freeway, relative to the direction of the road segments in the freeway.



5. Lay down a few more nodes to continue the ramp. Then connect the ramp back to the freeway by clicking a node on the freeway. When you do this, an intersection might initially form instead of the "merge" you want (see figure below).



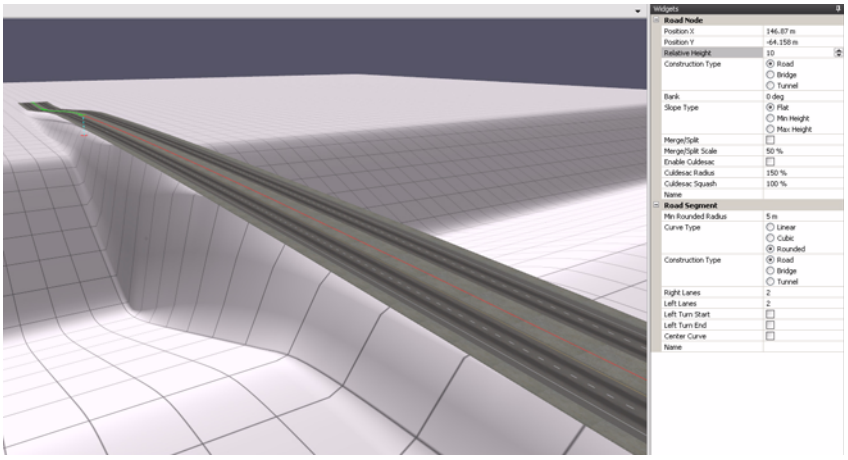
6. If an intersection forms, convert it to a merge: Select the **target node** in the intersection on the freeway, and then enable **Merge/Split** in the Road Node Widgets, as you did a few steps ago.

Build Bridges and Overpasses

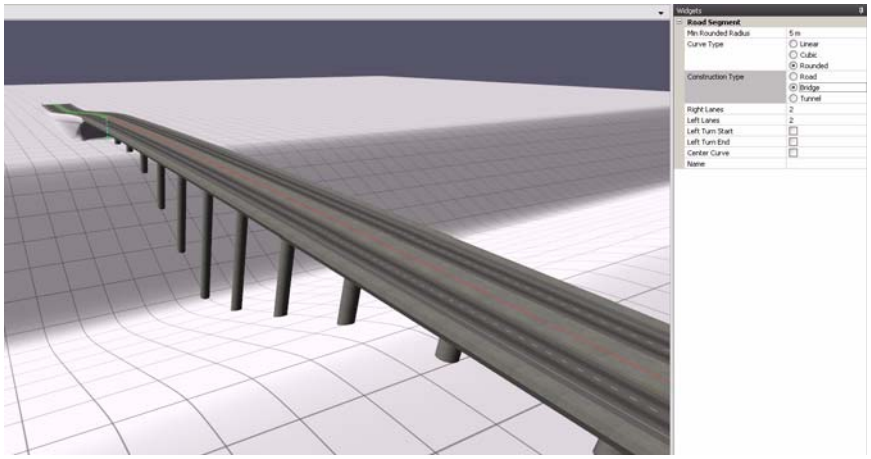
1. Start with the section of freeway, plus merge/split ramp, that you built in the previous section, Create Merges and Splits.

2. Select all the nodes and segments in the freeway, but do not select the nodes or segments unique to the ramp.

3. In the Road Node Widgets, change the *Construction Type* of the selected nodes to **Bridge**, and raise their *Absolute / Relative Height* to **10 meters**. This action elevates the road on a land fill, but does not add the pylons for the bridge until the road segments are also changed to the Bridge Construction Type.



4. In the Road Segment Widgets, change *Construction Type* of the selected segments to **Bridge**. The stretch of freeway becomes a long bridge elevated by pylons.



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More Information on Road Tools

Many additional features and subtleties within the road system are not covered in this short Quick Start. Examples include customizing road styles with imported textures, advanced editing of curves and tangents, creation of tunnels, and more. For details, see “Chapter 6: Using Road Tools” in the CityScape User Manual.

Using Water Tools

The Water Tools let you add global and/or local water to your terrains:

- Using The Global Water Tool, you can submerge everything in your terrain that’s lower in elevation than **Height** elevation in the Main Widgets. Global water is particularly useful for oceans and large lakes.
- Using The Water Polygon Tool, you can draw the shores of lakes at any elevation.

The Global Water Tool

Follow these steps to create and customize global water in your terrain:

1. With a terrain loaded in CityScape, activate the **Global Water Tool** by CLICKING its ICON in the Tools Palette.
2. To turn on water, enable the **Global Water** widget in the Global Water Widgets.
3. To set the elevation of the global water plane, adjust the **Height** widget. Try raising and lowering the water, and adjusting other water widgets.

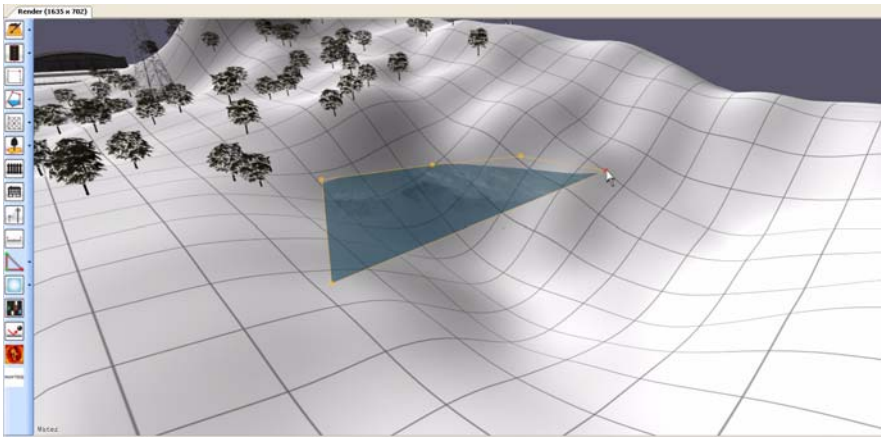
   Tip: *First create some valleys that drop below ground level, so the water has a place to pool. Then use a negative value in the Height widget so the water pools only in these valleys without flooding your entire terrain.*

The Water Polygon Tool

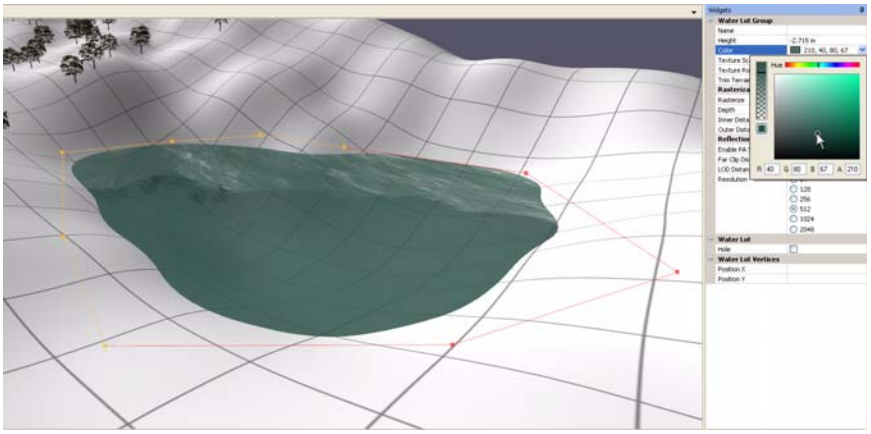
Using the Water Polygon Tool, you can draw the shores of lakes at any altitude in your terrain.

1. With a terrain loaded in CityScape, activate the **Water Polygon Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Water drop-down list to see the icon for the Water Polygon Tool.)

2. RIGHT-CLICK in the terrain a few times to define the shores of your lake. (Using the Water Polygon Tool is almost identical to using The Geography Polygon Tool, except water appears within the borders of water polygons, of course.)



3. To customize a Water Polygon, first select at least one point in the polygon, and then adjust the Main Widgets.

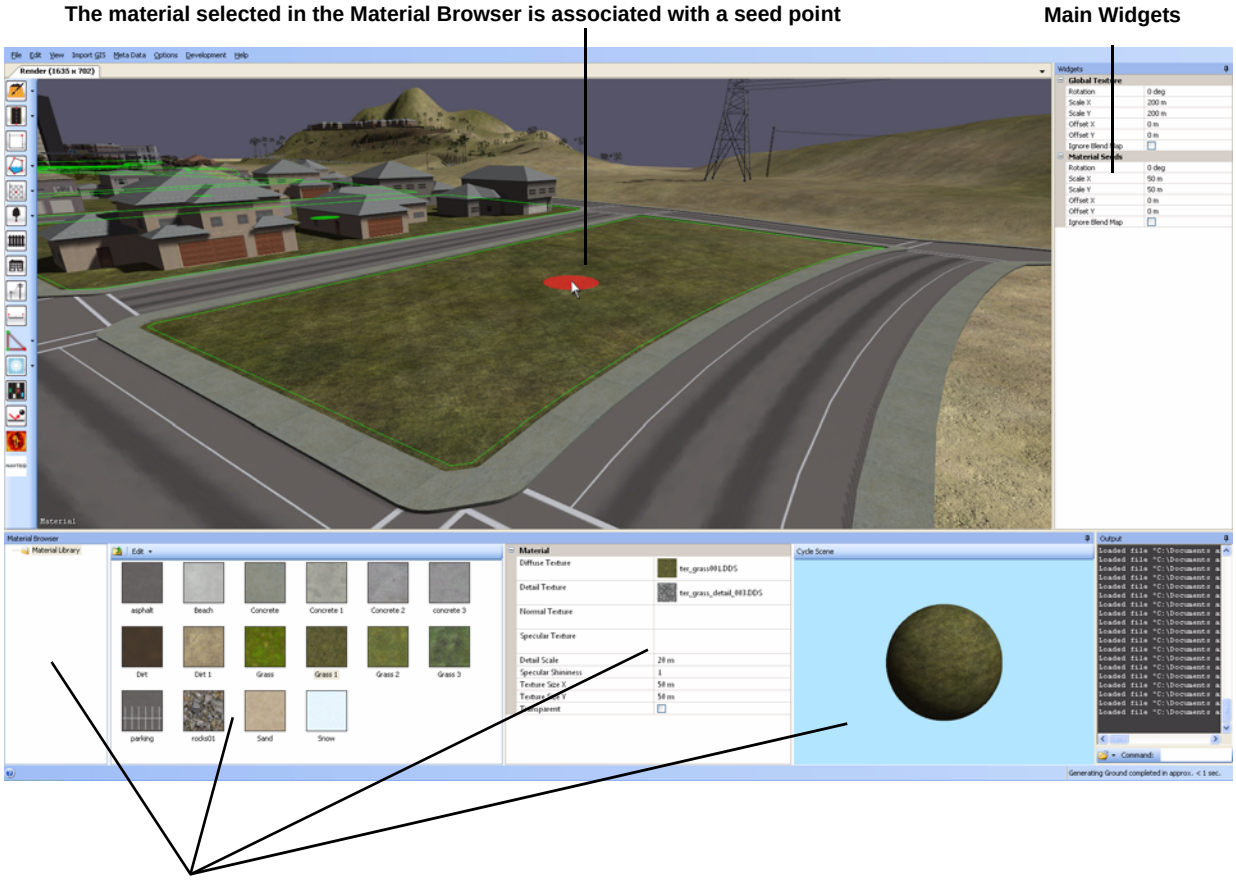


Using Material Tools

Using the Material Browser, you can assign a global material (such as dirt) to the entire terrain. Using the Material Seed Tool, you can assign a different a material (such as asphalt) to a "lot" or closed loop or road.

About the Material Browser

When you select any of the Material Tools from the Tools Palette, the Material Browser appears in the lower part of the CityScape window, by default (see figure below). You use the Material Browser to select materials to map onto geography; to create new materials; and to edit existing materials.



Material Browser showing (from left to right) the Material Tree, Material Icons, Material Sub-Widgets, and Material Preview

To select a material in the Material Library, simply LEFT-CLICK the ICON of the desired material in the Icon View of the Material Browser. A sample of the selected material will appear in the Preview Window.

About Material Sub-Widgets

As with Road Style Sub-Widgets, the Material Sub-Widgets allow you to edit the *global* properties of a material. Your edits will affect all instances of the material in the open terrain, as well as other terrains using the same material library.

The Material Seed Tool

You can use the Material Seed Tool to map any material in the Material Library to a “lot” in the open terrain. (A lot is any region completely enclosed by roads, or a lot polygon created with Lot Tools.) Materials assigned to lots override the global material.

1. A closed loop of road should already exist in your terrain as a result of Using Road Tools. If not, use the Road Tools to create it.

2. Activate the **Material Seed Tool** by CLICKING its ICON in the Tools Palette.

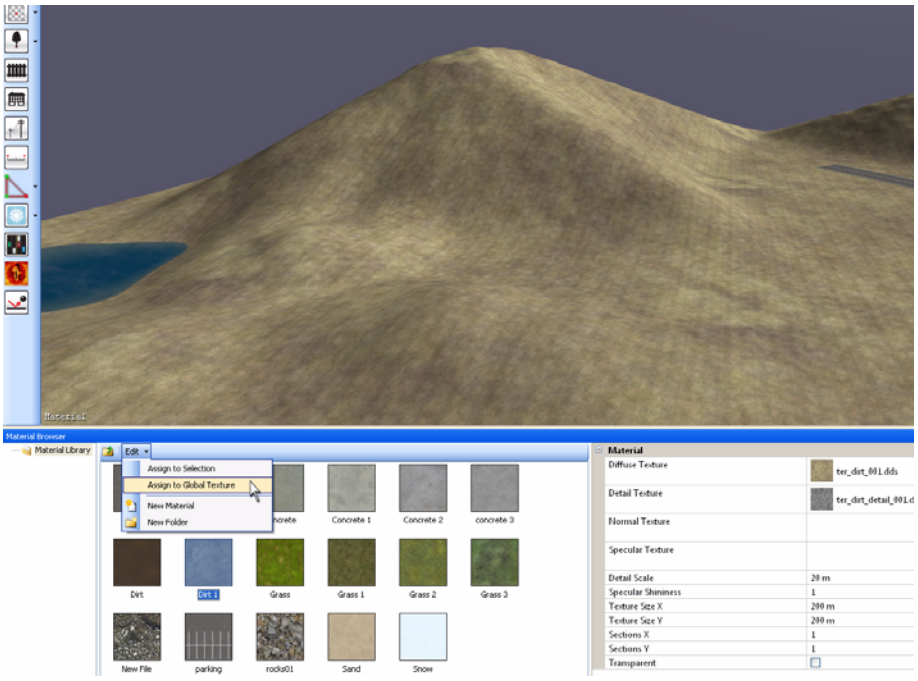
3. To assign a material to the “lot” formed by a closed loop of road, first select the material by LEFT-CLICKING the material’s **icon** in the Material Browser. Then RIGHT-CLICK anywhere in the **target lot**. This action places a material seed point within the lot, assigns the selected material to the seed point, and maps the material to the lot. For an example, see the image in About the Material Browser.

The Global Material Feature

When you map a global material to an entire terrain, the texture will appear on all geography (i.e. mountains, valleys, etc) – except for lots that have specific textures assigned to them. Textures assigned to lots override global textures.

1. Activate the **Material Seed Tool** by CLICKING its ICON in the Tools Palette.

2. Without placing any points on the terrain, LEFT-CLICK the icon of the desired Material in the Material Browser. Then select **Edit > Assign to Global Texture** in the Edit Menu of the Material Browser. (Shortcut: DOUBLE-CLICK the icon of the target material to use it globally in the open terrain.)



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Edit Materials

To modify a material, follow these steps:

1. With a terrain loaded in CityScape, activate the **Material Seed Tool**.

2. To select a different material for a lot that already has an assigned material, first LEFT-CLICK the **seed point** in the target lot. Then DOUBLE-LEFT-CLICK the **icon** of the desired material in the Material Browser.

3. To modify the local properties of a material used in a terrain, first LEFT-CLICK the seed point of the material to select it, and then edit its parameters in the *Main Widgets*. Experiment with these widgets.



4. To modify the global properties of a material, LEFT-CLICK the **icon** of the target material in the Material Browser, and edit its parameters in the Material Sub-Widgets. Among other things, you can import textures that get blended to create a complete material.

Using Prop Tools

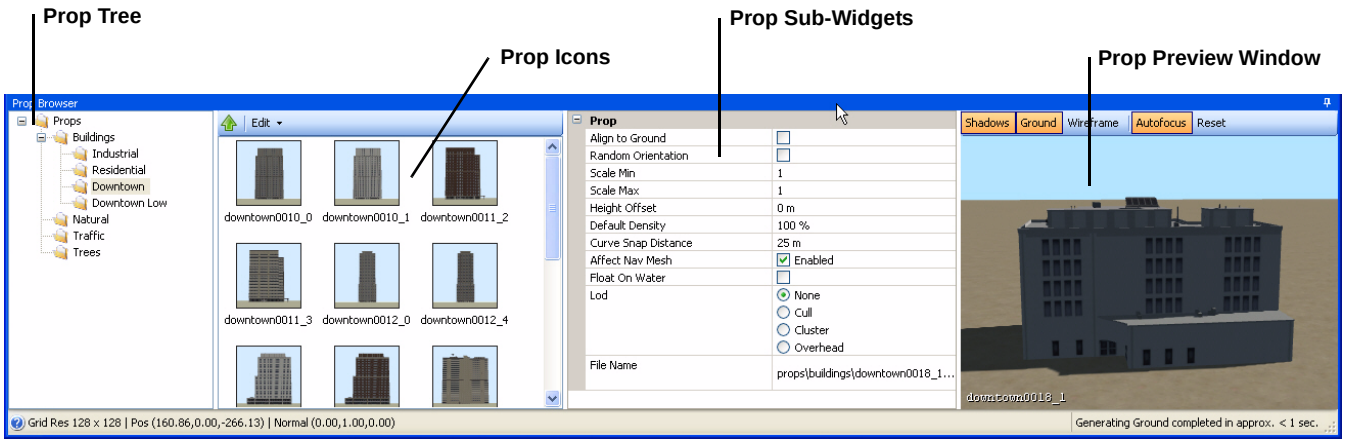
Props are 3D models that you can create and texture in external software such as 3D Studio Max, Creator, and Maya. Examples of props include buildings, trees, mailboxes, street lights, cars, and so forth. Using the Prop Tool Group, you can place props into a terrain and modify their properties.

About The Prop Browser

The Demo Project comes with a Prop Library that includes some buildings, cars, trees, etc. Using the Prop Browser in CityScape, you can select the props in the Prop Library; import, manage, and modify props; and select props to place into terrains.

1. With a terrain loaded in CityScape, activate the **Prop Point Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Point Tool). The Prop Browser appears along the bottom of the window.

2. Using the Prop Browser, explore the contents of the Prop Library in the Demo Project. When you click an icon in the Prop Icons, the Prop Sub-Widgets show its global properties (i.e. the properties shared by all instances of the prop). You will also see a sample of the prop in the Prop Preview Window. See the figure below for an example of the Prop Browser and its panels.



IMPORTING NEW PROPS

With the Full version of CityScape, you can import, manage, and edit new props, as well as save libraries and terrains. (The Demo Version does not allow you to save or export terrains or libraries.) This section gives you basic instructions for importing a prop. For simplicity, these instructions do not document all the secondary factors that pertain to importing complex assets with multiple LODs. For details on these factors, see “Chapter 10: Using Prop Tools” in the CityScape User Manual.

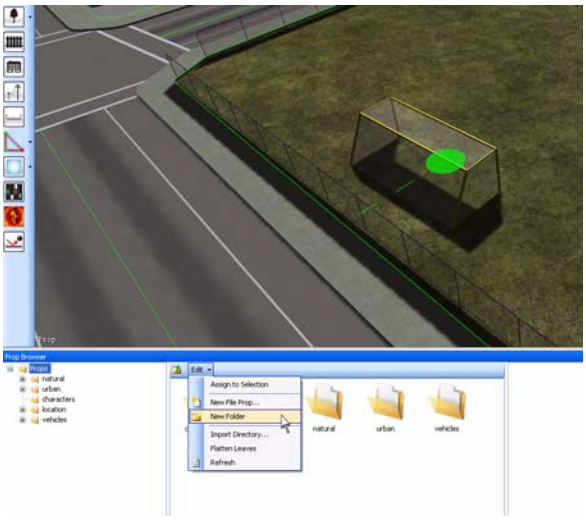
1. Export your prop from the external modeling software that created it.

   Note: *The **exported prop** should be a 3D model in Collada (DAE) format. If the prop includes textures, these textures need to be in a common 2D format such as JPG, and applied to the prop in your external modeling software. When exported from Maya or other creation software, the Collada prop file (DAE format) and the textures need to be exported to the same folder.*

2. Move the exported Collada file for the new prop, together with its textures, into a folder under the project file (CSP) associated with your target CityScape terrains.

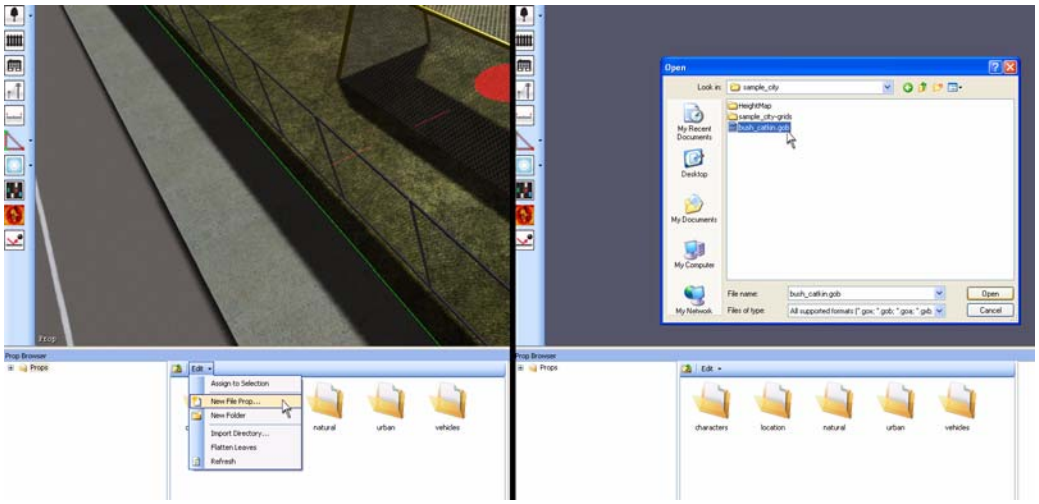
3. With a terrain loaded in CityScape, active the **Prop Point Tool** by CLICKING its ICON in the Tools Palette. The Prop Browser appears near the bottom of the screen, by default.

4. In the Prop Tree of the Prop Browser, LEFT-CLICK any **target folder** to contain the new prop, or create a new folder.



   – Note: The folders in the Prop Tree do NOT correspond to the folders on your hard drive, but using the same folder names and tree structure are good ways for you to avoid confusion.

5. Along the top of the Prop Icons Window in the Prop Browser, LEFT-CLICK **Edit > New File Prop**. The icon for a new blank prop appears in the Prop Icon window, and a file browser appears.



6. In the file browser, navigate to the **model file** (DAE) of your target prop and open it. The new prop and its textures will be converted to PixelActive formats and added to the Prop Library in the current project.

Once you've added a prop to the Prop Library, you can place the new prop into any terrain that references the same project file (CSP).

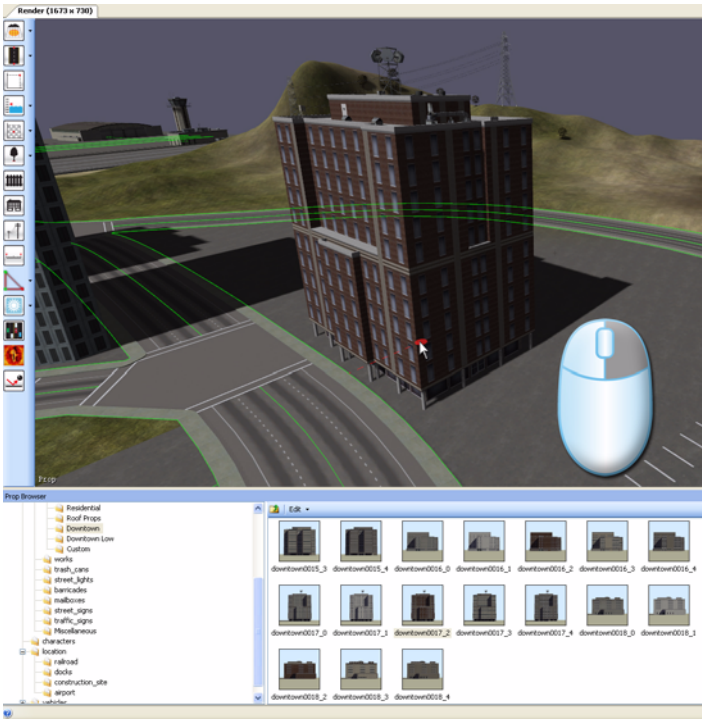
THE PROP POINT TOOL

Place individual props using the *Point Placement Tool*:

1. With a terrain loaded in CityScape, active the **Prop Point Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Point Tool).

2. In the Prop Browser, use the Prop Tree and Prop Icons to locate one of the buildings in the Prop Library. LEFT-CLICK the icon the desired building to select it.

3. To place the selected prop in a terrain, RIGHT-CLICK the target location in the terrain.



Before moving a prop point that already exists in the terrain, or editing its properties using the Main Widgets, you must first select it. To select a Prop Point in a terrain, hover the mouse pointer over the **placement point** of the prop, and then LEFT-CLICK-HOLD and DRAG the MOUSE.

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The Prop Circle Tool

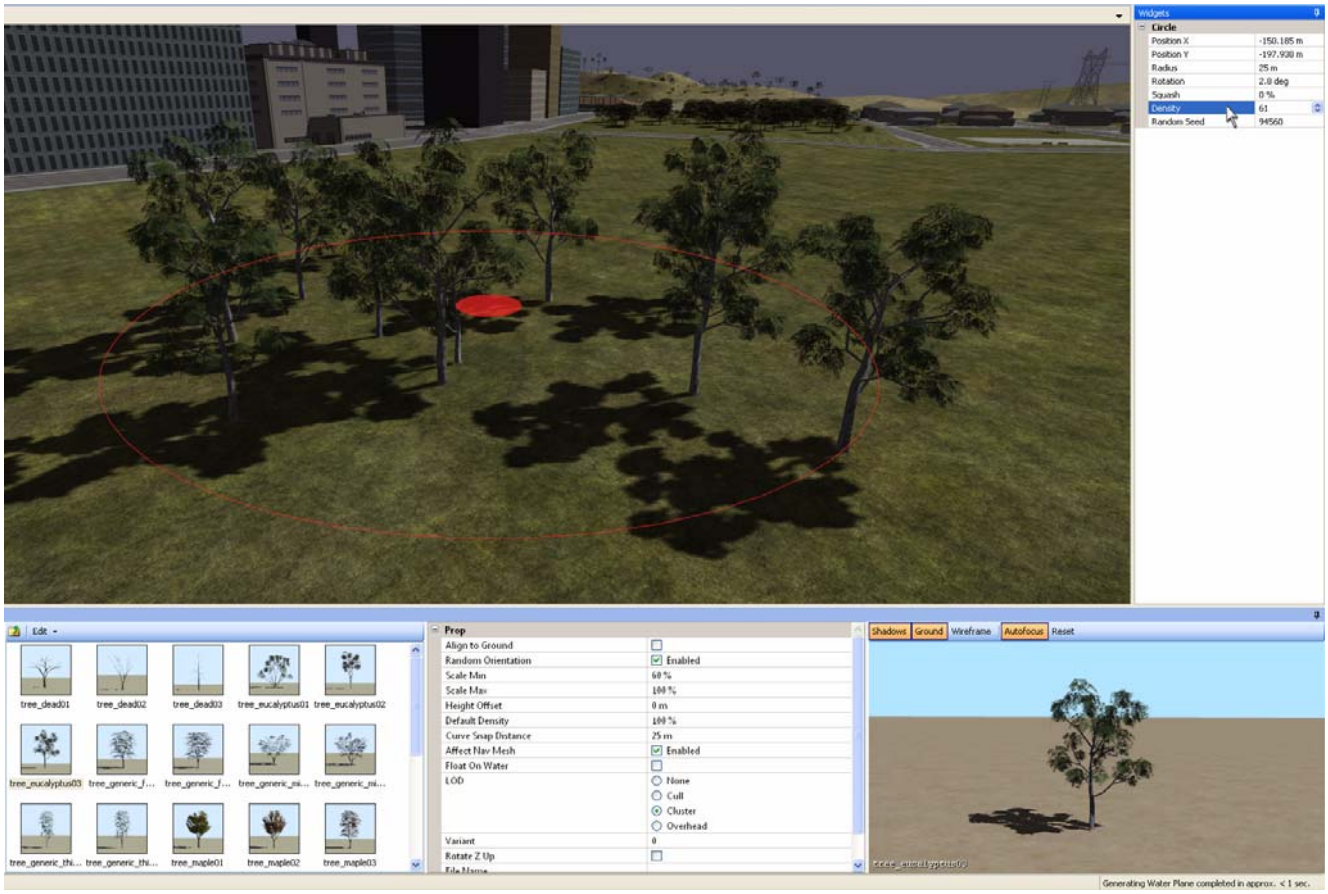
The Prop Circle Tool lets you distribute props in Circle Features, as described below:

1. With a terrain loaded in CityScape, activate the **Prop Circle Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Circle Tool).

2. Use the Prop Browser to locate and select any tree from the Prop Library.

3. RIGHT-CLICK in the terrain to place a Circle Feature in nearly the same way you did with the The Geography Circle Tool. Multiple instances of the tree prop appear within the Circle Feature.

4. LEFT-CLICK the center of the Prop Circle Feature to actively select it, and modify its parameters in the Main Widgets.



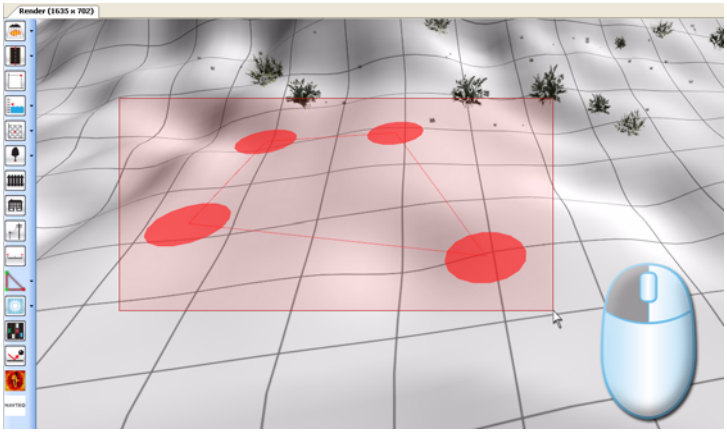
THE PROP POLYGON TOOL

The Prop Polygon Tool lets you distribute props in Polygon Features. Here's how you use this tool:

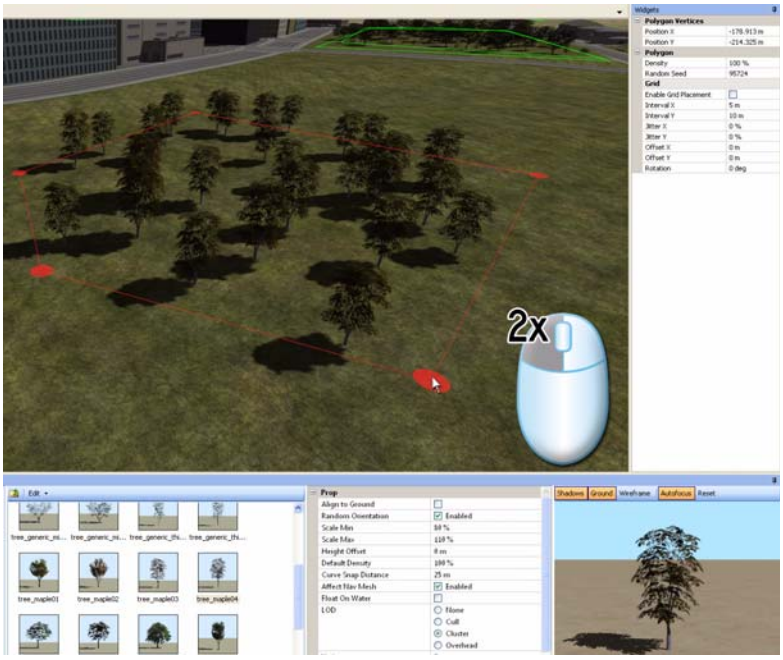
1. With a terrain loaded in CityScape, active the **Prop Polygon Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Polygon Tool).

2. Using the Prop Browser, locate and select any tree from the Prop Library.

3. Place points defining a polygon in the terrain, just as you would with The Geography Polygon Tool. Multiple instances of the selected prop appear within the polygon.



4. To select the entire Prop Polygon, DOUBLE-LEFT-CLICK any point in it. You can use the Main Widgets to modify the local parameters of the actively selected Prop Polygon, and use Prop Sub-Widgets to modify the global parameters of the props within in the Prop Polygon.

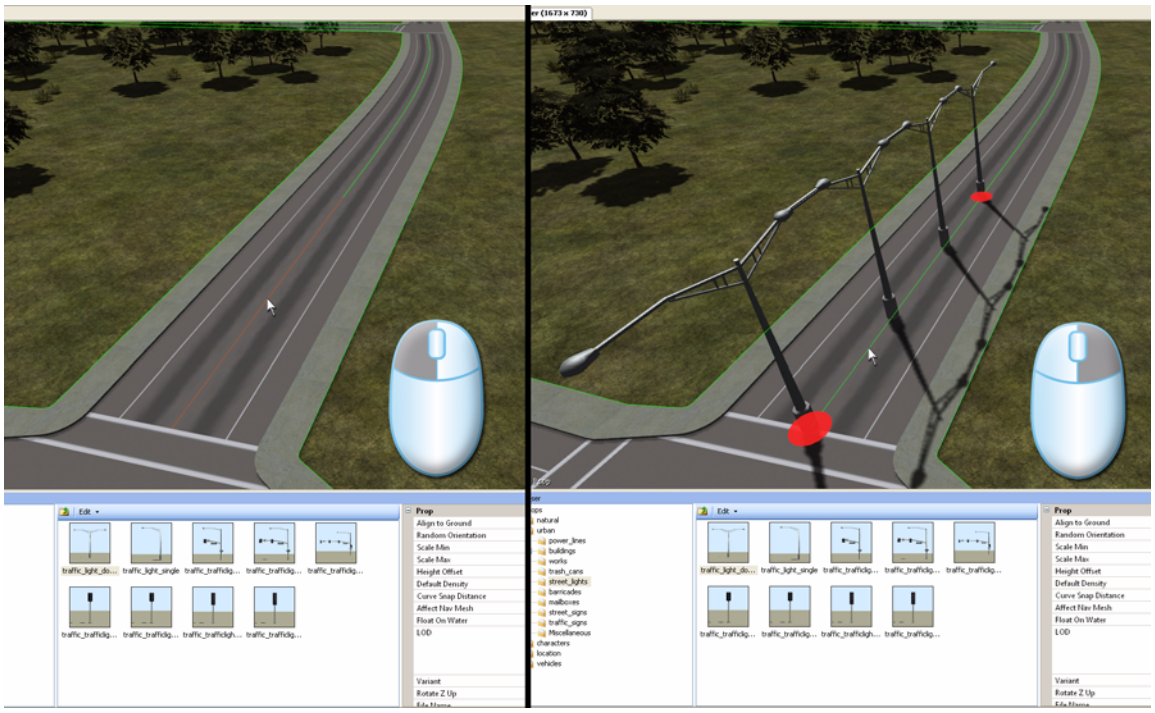


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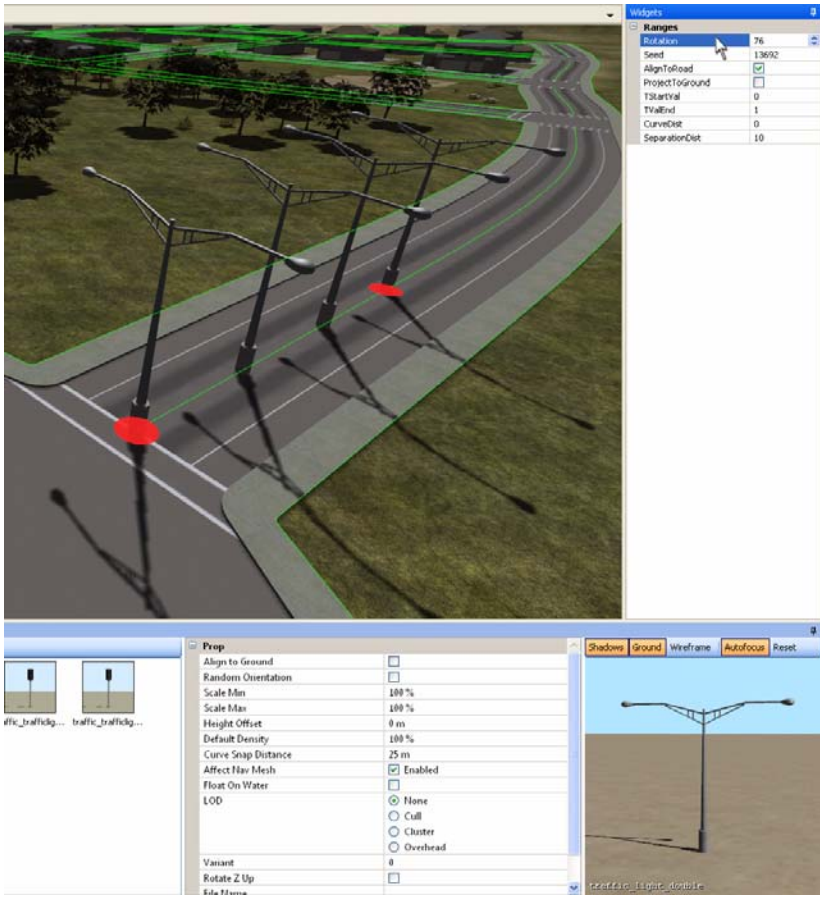
THE PROP RANGE TOOL

The Prop Range Tool is most useful for repeating props, such as trees or street lights, along the sides of roads.

1. Ensure your CityScape terrain contains at least one **road segment**.
2. Active the **Prop Range Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Prop drop-down list in the Tools Palette to see the Prop Range Tool).
3. In the Prop Browser, select any street light from the Prop Library.
4. LEFT-CLICK a road segment to select it, and then RIGHT-CLICK that same segment. The prop will appear repeatedly from the start to end nodes of the selected road segment.



5. To select the entire Prop Range Feature, LEFT-CLICK the start or end node of the road segment associated with the prop range. Use the Main Widgets to modify the local parameters of a Prop Range, and use Prop Sub-Widgets to modify the global parameters of the prop used in the Prop Range.



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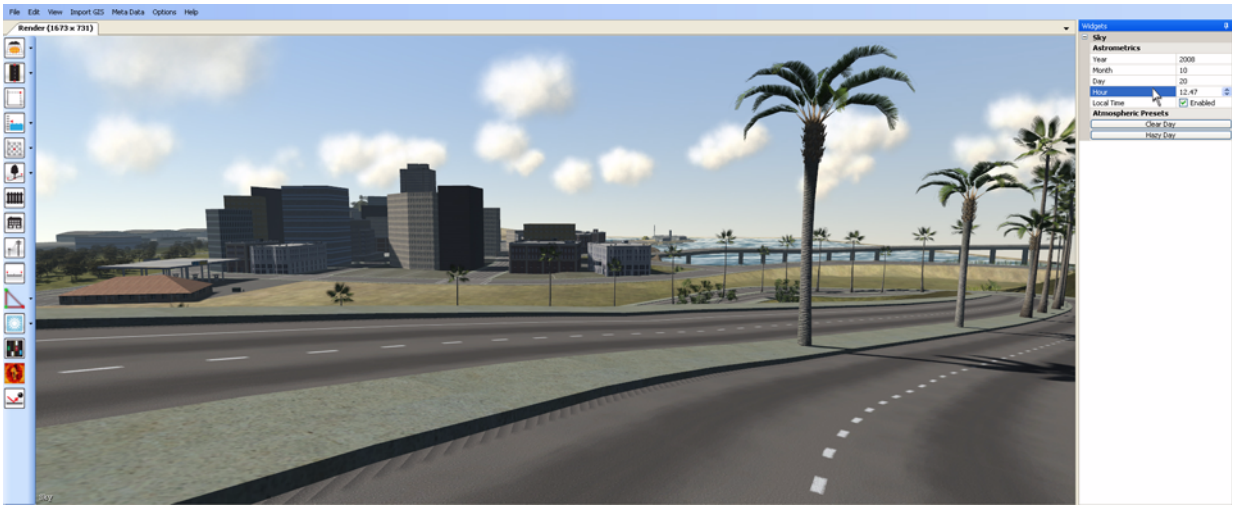
Using Sky Tools

Using the Sky Tools, you can edit atmospheric properties such as clouds and haze, and astronomical properties that affect natural lighting. Based on the date, time, and location you enter, CityScape will accurately determine the positions of the sun and moon, and illuminate the terrain accordingly.

The Sky Tool

1. Activate the **Sky Tool** by CLICKING its ICON in the Tools Palette.

2. Use the Main Widgets to customize the sky:
   – Specify date and time
   – Select between a clear or hazy day



If you want the settings in the Sky Widgets to position the sun accurately for a specific time, date, and location, the terrain must be geo-located using latitude and longitude. For details on geo-locating a new terrain, see Create a New Geo-Located Terrain. To geo-locate an existing terrain, follow the steps below:

1. Select **Options > Terrain Settings** in the Main Menu. The Terrain Settings dialog appears.

2. Fill in the **Geolocated Terrain** side of the Terrain Settings dialog just as you would if you were creating a new geo-located terrain.

For background information on geo-located terrains and related subjects, such as Shapefiles and elevation files, see “Chapter 21: Importing GIS Data” in the *CityScape User Manual*.

The Clouds Tool

1. Activate the **Clouds Tool** by CLICKING its ICON in the Tools Palette. (You may need to expand the Sky drop-down list in the Tools Palette to see the Clouds Tool).

2. Experiment with the Main Widgets to customize the sky.

Using GIS Tools

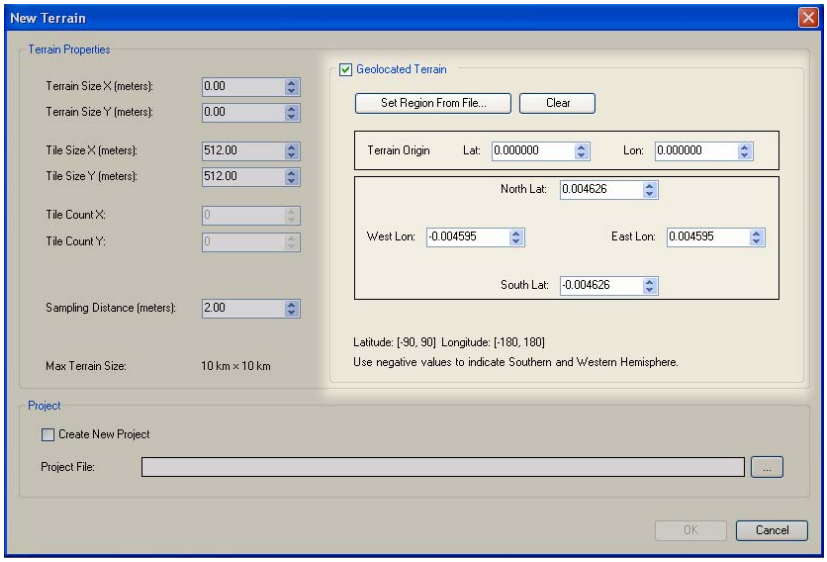
This section provides you with a simple case of importing GIS data into CityScape in the following phases:

1. Create a New Geo-Located Terrain
2. Import Elevation Data
3. Import Roads, Buildings, Water, and Props
4. Import an Orthographic Image

If you are unfamiliar with geo-specific data, read the background information in “Chapter 21: Importing GIS Data” in the *CityScape User Manual*.

Create a New Geo-Located Terrain

1. PRESS **Ctrl-N** to create a new terrain. The New Terrain dialog appears.
2. In the New Terrain dialog, LEFT-CLICK the **Geolocated Terrain** check box to enable the right side of the dialog, which pertains to the geo-location of the new terrain.



3. Select the **Demo Project** by LEFT-CLICKING the **Project File [...]** button near the bottom, and loading the target **demo.csp** file located in the Worlds folder, which is installed here by default:

C:\Users\Public\Documents\PixelActive\CityScape 1.8 \Worlds

or here for WinXP users:

C:\Documents and Settings\All Users\Documents\PixelActive\CityScape 1.8\Worlds

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4. Set the geo-location of the new terrain as follows:
   i. LEFT-CLICK the **Set Region from File** button.
   ii. In the dialog that appears, set *Files of Type* to **All Files**. Then locate and load the **lajolla_height.tif** elevation file in the *Sample GIS* folder under the main *Worlds* folder (see previous step).

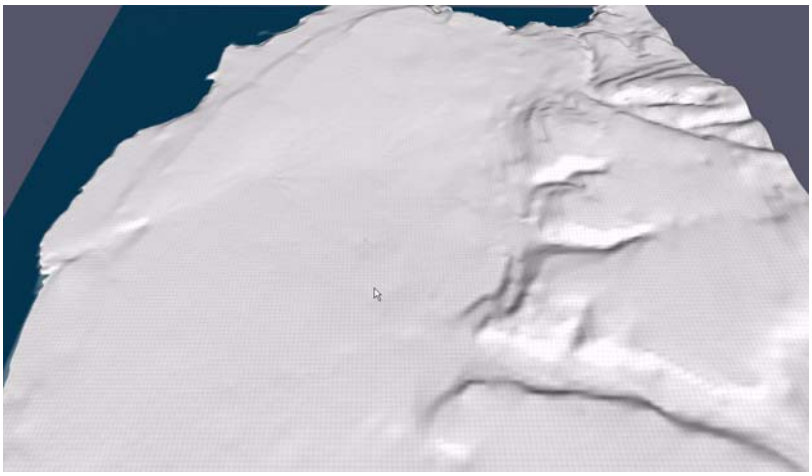
5. LEFT-CLICK **OK** to create a blank terrain that's geo-located. You can now import geo-specific data into this terrain.

Import Elevation Data

In a typical workflow, you import elevation files before importing any other types of geo-data:

6. To import an elevation file, select **Import GIS > Import Elevation > GeoTiff** from the Main Menu. (Note: CityScape can also import elevation files in DEM, DTED, and GridFloat formats).

7. In the Import Elevation file loader, locate and load the same **lajolla_height.tif** elevation file you just used to geo-locate the terrain. The geography in the terrain will be updated to match the height field you just imported.



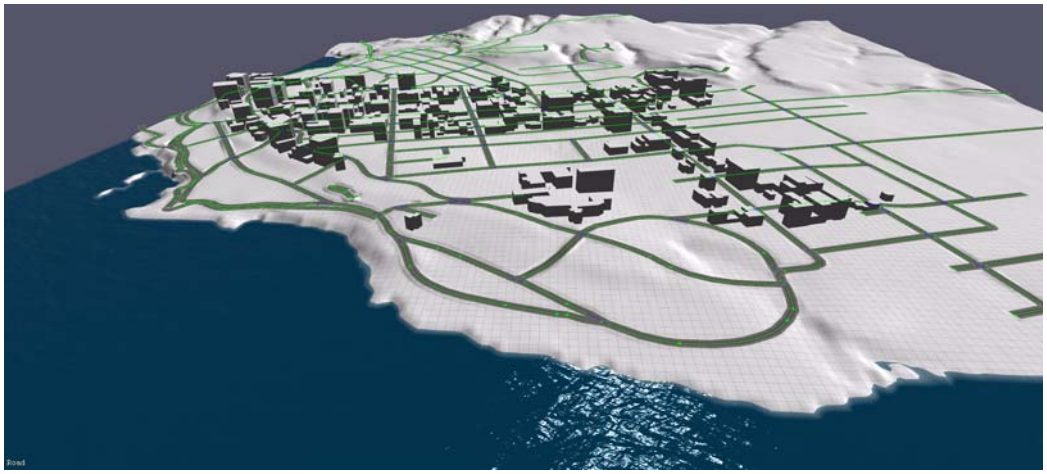
Import Roads, Buildings, Water, and Props

GIS data for roads, buildings, water, and props in the same region would require 12 different Shapefiles, or in one OpenStreetMap file. For simplicity, these instructions describe how to import an OpenStreetMap file:

8. To import an OSM data file, select **Import GIS > Import OpenStreetMap** from the Main Menu. The *Import OpenStreetMap* dialog appears.

9. In the *OpenStreetMap File* tab of the dialog:
   – Locate and load the **lajolla.osm** file in the same folder as the elevation file you just loaded.
   – Select the *Features to Import* (normally all).

10. In the *OpenStreetMap File* tab, LEFT-CLICK **OK** to finish the import.



IMPORT AN ORTHOGRAPHIC IMAGE

You can optionally import orthographic imagery in the same region, so as to compare a satellite photo with the 3D data you just imported:

11. Select **Import GIS > Import OrthoImagery** from the Main Menu. The *Import Ortho Image* dialog appears.

12. In the dialog, locate and load the **lajolla_ortho.tif** orthographic image file. The ortho image overlays your terrain. You can show or hide the ortho image with the **View > Show Overlay** toggle in the Main Menu.

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